

Spacefield Transformation I: A Geometric Closure Framework for Cosmological Mass, Confinement Scale, and Expansion Dynamics

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Abstract

Within the minimal SFT closure framework, the global mass–curvature configuration is constrained by a compactness-saturation condition and a projected mass–area partition. The dimensionless closure structure determines the relative confinement geometry, while the absolute cosmological scale is specified by the baseline confinement normalization

$$\eta_{\min} = \frac{2}{9} \text{ kg m}^{-2}.$$

Conditional on this minimal SFT normalization, the framework yields

$$M_m \simeq 9.740 \times 10^{53} \text{ kg}, \quad R_c \simeq 1.4466 \times 10^{27} \text{ m}.$$

These quantities define the baseline mass and confinement scales used for the subsequent reconstruction of the late-time cosmological background.

Observed cosmological energy fractions are then used to reconstruct the present cosmological background associated with the closure state. The framework yields a Hubble scale and dark-energy density consistent with current observational constraints, together with the transport law

$$\rho_{\text{DE}}(a) = \rho_{\text{DE},0} a^{3(1-\alpha)},$$

which reproduces Λ CDM in the limit $\alpha = 1$ while allowing small observationally testable deviations for $\alpha \neq 1$.

For representative departures $|\alpha - 1| \sim 10^{-2}$, the parametrised background produces sub-percent to percent-level modifications to late-time expansion and cosmological distance observables relative to Λ CDM. These calculations establish conditional observational signatures through which the proposed boundary-scaling sector may be constrained, including independent tests with electromagnetic distances and gravitational-wave standard sirens.

Symbols and Definitions (Cosmological Sector)

Symbol	Definition / Meaning / Units (and value if given)
Fundamental Geometry and Closure Relations	
UCZ	Universal Confinement Zone; finite cosmological closure domain defined within the SFT framework.
χ	Dimensionless compactness parameter defined by $\frac{2GM_m}{R_c c^2} = \chi,$ with the minimal SFT closure condition corresponding to $\chi = 1$.
$\mathcal{C}(R)$	Dimensionless compactness functional, $\mathcal{C}(R) = \frac{2GM}{Rc^2}.$
$\mathcal{F}[R, M]$	Compactness-mismatch functional, $\mathcal{F}[R, M] = \left(R - \frac{2GM}{c^2}\right)^2.$
M_{MS}	Misner–Sharp quasi-local mass associated with a spherically symmetric domain.
Fixed Geometric Invariants (SFT Baseline)	
M_m	Total comoving matter rest mass confined within the Universal Confinement Zone, $M_m = \frac{3c^4}{8\pi G^2 \eta_{\text{min}}} = 9.74014 \times 10^{53} \text{ kg}.$
R_c	Universal Confinement Zone radius defined by curvature saturation, $R_c = \frac{2GM_m}{c^2} = 1.4466 \times 10^{27} \text{ m}.$
M_α	Effective unconfined or leakage-sector mass component, $M_\alpha = \frac{2}{3}M_m = 6.49343 \times 10^{53} \text{ kg}.$

Symbol	Definition / Meaning / Units (and value if given)
$\eta_{\min}$	Baseline dimensional confinement normalization of the minimal SFT closure branch, $\eta_{\min} = \frac{2}{9} \text{ kg m}^{-2}.$
H_{∞}	The dimensionless SFT mass–area partitions determine relative closure ratios, while $\eta_{\min}$ supplies the absolute surface-density scale. Asymptotic expansion scale associated with the UCZ radius, $H_{\infty} = \frac{c}{R_c} = 2.072 \times 10^{-19} \text{ s}^{-1} \approx 6.39 \text{ km s}^{-1} \text{ Mpc}^{-1}.$
Present-Epoch Cosmological Quantities	
R_o	Present comoving particle-horizon radius of the observable Universe, $R_o = 4.39448 \times 10^{26} \text{ m}.$
V_0	Present observable comoving volume, $V_0 = \frac{4\pi}{3} R_o^3 = 3.55477 \times 10^{80} \text{ m}^3.$
$M_{r,0}$	Present radiation mass enclosed within R_o , $M_{r,0} = 2.79344 \times 10^{50} \text{ kg}.$
$M_{\text{DE}}(t_0)$	Present dark-energy mass-equivalent enclosed within R_o , $M_{\text{DE}}(t_0) = 2.11778 \times 10^{54} \text{ kg}.$
$M_u(t_0)$	Total mass–energy enclosed within R_o , $M_u(t_0) = 3.09210 \times 10^{54} \text{ kg}.$
H_0	Present-day Hubble constant. For the baseline SFT closure-consistency reconstruction, $H_0 = 68.05 \text{ km s}^{-1} \text{ Mpc}^{-1}.$ This value is reconstructed using the closure-normalized mass scale and the adopted cosmological distance and density anchors.

Symbol	Definition / Meaning / Units (and value if given)
t_0	Present cosmic age, $t_0 \simeq 13.66$ Gyr.
R_∞	Asymptotic future observable radius predicted by the SFT closure system, $R_\infty \approx 18\text{--}20$ Gpc.
Density Parameters and Energy Densities	
ρ_c	Standard FRW critical density, $\rho_c = \frac{3H_0^2}{8\pi G}.$
$\rho_{m,0}$	Present matter density, $\rho_{m,0} = \Omega_{m,0}\rho_c.$
$\rho_{r,0}$ $\Omega_{m,0}$	Present radiation density. Present matter density fraction, $\Omega_{m,0} = 0.31500.$
$\Omega_{r,0}$	Present radiation density fraction, $\Omega_{r,0} = 9.03400 \times 10^{-5}.$
$\Omega_{\text{DE},0}$	Present dark-energy density fraction, $\Omega_{\text{DE},0} = 0.68491.$
$\rho_{\text{DE},0}$	Present dark-energy mass density, $\rho_{\text{DE},0} = 5.95765 \times 10^{-27} \text{ kg m}^{-3}.$

Symbol	Definition / Meaning / Units (and value if given)
$u_{\text{DE},0}$	Present dark-energy energy density, $u_{\text{DE},0} = 5.35447 \times 10^{-10} \text{ J m}^{-3}.$
$\Omega_i(z)$	Fractional density of cosmological component i as a function of redshift z .
SFT Dynamics and Evolution Laws	
$a(t)$	Cosmological scale factor.
z	Cosmological redshift, $a = (1 + z)^{-1}.$
$H(z)$	Cosmological expansion rate as a function of redshift.
$E(a)$	Normalized expansion function, $E(a) = \frac{H(a)}{H_0}.$
$\rho_{\text{DE}}(a)$	Dark-energy density evolution law, $\rho_{\text{DE}}(a) = \rho_{\text{DE},0} a^{3(1-\alpha)}.$
α	Effective dark-energy evolution index used in the SFT boundary-scaling parametrisation, $\rho_{\text{DE}}(a) = \rho_{\text{DE},0} a^{3(1-\alpha)}.$ Within Paper I, α is not assigned a unique SFT-predicted numerical value. The limit $\alpha = 1$ reproduces constant dark energy and the Λ CDM background.
p	Phenomenological boundary-transport exponent of the minimal SFT dark-energy ansatz. It is related to α by $p - 1 = 3(1 - \alpha), \quad \alpha = 1 - \frac{p - 1}{3}.$ Paper I does not derive a unique value of p from the UCZ closure geometry.
D_{eff}	Effective transport dimension, $D_{\text{eff}} = 2 + p.$

Symbol	Definition / Meaning / Units (and value if given)
w_{eff}	Effective constant equation-of-state parameter associated with the boundary-scaling law, $w_{\text{eff}} = \alpha - 2.$ Thus $\alpha = 1$ corresponds to $w_{\text{eff}} = -1$, $\alpha < 1$ to $w_{\text{eff}} < -1$, and $\alpha > 1$ to $w_{\text{eff}} > -1$.
$\dot{M}_{\text{DE}}$	Instantaneous dark-energy mass-growth rate, $\dot{M}_{\text{DE}} = (4.4 \times 10^{44})(1 - \alpha) \text{ kg yr}^{-1}.$
$\Delta H/H$	Fractional expansion-rate deviation between SFT and Λ CDM.
$\Delta\mu(z)$	Distance-modulus residual between SFT and Λ CDM supernova Hubble diagrams.
$\Delta_{\text{siren}}(z)$	Conditional standard-siren luminosity-distance residual between the SFT and Λ CDM backgrounds, $\Delta_{\text{siren}}(z) = \frac{d_L^{\text{SFT}}(z)}{d_L^{\Lambda\text{CDM}}(z)} - 1.$

Fundamental Constants

c	Speed of light in vacuum.
G	Newton gravitational constant.

Contributions. This paper develops a geometric closure construction for cosmological observables within the Spacefield Transformation (SFT) and Universal Confinement Zone (UCZ) framework:

1. Derive closed forms for the total matter rest mass M_m and confinement radius R_c conditional on the UCZ closure relation and the specified dimensional confinement normalization η_{min} ;
2. Establish the 1:2:3 mass and 1:3:4 partition structure of the minimal SFT closure construction;
3. Combine the resulting closure-normalized matter mass with observationally anchored cosmological fractions to reconstruct the present-epoch mass partition;
4. Reconstruct the corresponding dark-energy density and Hubble scale associated with the closure state;
5. Quantify conditional SFT departures from Λ CDM in $\Delta\mu(z)$, $H(z)$, and the standard-siren distance residual $\Delta_{\text{siren}}(z)$, while retaining standard General-Relativistic gravitational-wave propagation within each cosmological background.

Full derivations appear in the Appendix. The quantities M_m and R_c are derived conditionally within the minimal SFT closure framework from the UCZ compactness relation, the SFT mass–area partition structure, and the specified dimensional baseline normalization $\eta_{\min}$. Present-epoch cosmological fractions are subsequently used as observational anchors for reconstruction of the late-time background. We use M_u exclusively for the total mass within R_o , and R_c exclusively for the UCZ radius.

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Sections Overview

- **1. Introduction** — Motivation, limits of Λ CDM, and overview of the Spacefield Transformation (SFT) and Universal Confinement Zone (UCZ).
- **2. Foundational Geometry** — Geometric postulates, derivation of M_m and R_c , curvature closure, and confinement relations.
- **3. Cosmological Framework** — Determination of global observables (H_0 , t_0 , R_o/R_c), dark-energy law $\rho_{\text{DE}}(a) = \rho_{\text{DE},0}a^{3(1-\alpha)}$, and link to Λ CDM.
- **4. Predictions and Observables** — Testable and conditional observational outcomes involving H_0 , $H_\infty = c/R_c$, Ω_{DE} evolution, R_o/R_c , t_0 , $\Delta\mu(z)$, and standard-siren distance residuals.
- **5. Numerical Implementation and Results** — Updated 2025 parameters, computational methods, and Planck/DESI comparison.
 - **Table 1:** Fractional energy densities vs. redshift.
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 - **Figures 1–5:** Cosmic-age integrands, SFT- Λ CDM supernova residuals, fractional-density evolution, expansion-rate deviations, and standard-siren distance residuals.
- **6. Discussion and Outlook** — Physical interpretation and observational tests (LSST, Roman, Euclid, DESI, ET, LISA).
- **Appendix** — Derivations of the confinement-density normalization, mass partition, Hubble-scale relation, and supplementary figures and numerical diagnostics.

1 Introduction

A central objective of modern cosmology is to determine whether the global properties of the Universe, such as its total matter content, characteristic length scales, and expansion history, can be related to underlying geometric consistency conditions rather than introduced as independently calibrated parameters. The standard Λ CDM model provides a highly successful phenomenological description of a broad range of cosmological observations, including the cosmic microwave background (CMB), baryon acoustic oscillations, and Type Ia supernovae

[1–5]. Nevertheless, key quantities such as the total matter content, the present Hubble scale, and the physical origin of dark energy remain external inputs rather than derived consequences of the framework [6–8].

In this work we investigate the *Spacefield Transformation* (SFT), a geometric framework in which large-scale cosmological structure is constrained by a global confinement condition. The central idea is that the observable Universe may correspond to a finite effective domain whose boundary is selected by marginal compactness saturation [9, 10]. In this picture, the global curvature radius generated by the enclosed matter content coincides with the confinement radius of the domain itself.

This domain is referred to as the *Universal Confinement Zone* (UCZ). In the minimal SFT framework, the confinement geometry satisfies the global closure condition

$$R_c = \frac{2GM_m}{c^2}, \quad (1)$$

where M_m denotes the total conserved comoving matter content enclosed within the domain and R_c is the associated confinement radius.

Within SFT, Eq. (1) is interpreted not as a local Schwarzschild black-hole condition, but as a global compactness saturation relation for a finite self-gravitating cosmological domain [9, 10]. The compactness value

$$\frac{2GM_m}{R_c c^2} = 1$$

is adopted as the marginal closure condition of the minimal SFT branch. The compactness ratio alone is not used to classify the UCZ as a black-hole horizon or to determine its trapping character; such a classification would require the associated null expansions and full dynamical spacetime geometry.

A key feature of the framework is that the confinement state introduces a projected surface-density normalization,

$$\eta_{\min},$$

which fixes the global confinement geometry. Combining this confinement normalization with Eq. (1) yields the geometric relations

$$R_c = \frac{3c^2}{4\pi G\eta_{\min}}, \quad (2)$$

and

$$M_m = \frac{3c^4}{8\pi G^2\eta_{\min}}. \quad (3)$$

For the baseline dimensional confinement normalization of the minimal SFT branch,

$$\eta_{\min} = \frac{2}{9} \text{ kg m}^{-2},$$

the framework yields

$$M_m \simeq 9.740 \times 10^{53} \text{ kg}, \quad R_c \simeq 1.4466 \times 10^{27} \text{ m}.$$

These quantities are not fitted directly to late-time cosmological expansion data. Rather, they are derived conditionally within the minimal SFT model from the UCZ compactness relation, the SFT mass–area partition structure, and the specified dimensional normalization $\eta_{\min}$. The dimensionless closure geometry determines the relative structure of the construction, while $\eta_{\min}$ supplies its absolute surface-density scale.

Observed cosmological energy fractions are subsequently used to reconstruct the present cosmological background associated with this baseline closure state.

The SFT framework is constructed within the standard Friedmann–Robertson–Walker (FRW) background [11–13], with expansion dynamics governed by

$$H^2 = \frac{8\pi G}{3}\rho. \quad (4)$$

The total energy density is decomposed into matter, radiation, and dark-energy components. Observationally constrained quantities, including the present matter and radiation density fractions, $\Omega_{m,0}$ and $\Omega_{r,0}$, together with the observable radius R_o , are treated as external cosmological inputs used to reconstruct the late-time background associated with the SFT closure geometry.

To investigate possible departures from constant dark energy, the present paper introduces the phenomenological SFT boundary-scaling parametrisation

$$\rho_{\text{DE}}(a) = \rho_{\text{DE},0} a^{3(1-\alpha)}, \quad (5)$$

where α characterizes the effective dark-energy evolution.

The limit $\alpha = 1$ reproduces the constant-density Λ CDM background, while $\alpha \neq 1$ provides a controlled parametrisation of slowly evolving dark energy. Paper I does not derive a unique numerical value of α from the UCZ closure relation; rather, it examines the observational consequences of representative departures from $\alpha = 1$.

The purpose of the present paper is therefore twofold: first, to formulate the confinement-closure structure associated with the UCZ geometry; and second, to investigate whether the resulting framework yields a self-consistent and observationally testable cosmology when embedded within the standard FRW background.

Specifically, we:

1. formulate the UCZ relation as a global confinement condition;
2. determine the associated confinement radius and matter scale conditional on the specified SFT dimensional normalization;
3. reconstruct the corresponding late-time cosmological background;
4. formulate a phenomenological boundary-scaling parametrisation for the dark-energy evolution and examine its observational consequences;
5. evaluate observational signatures in $H(z)$, supernova distance moduli, and standard-siren luminosity-distance residuals relative to Λ CDM.

These results provide a basis for assessing whether global confinement geometry can serve as a viable extension or reinterpretation of standard cosmology.

2 Variational Motivation and Consistency with General Relativity

The Spacefield Transformation framework can be interpreted as a geometric closure condition applied to a standard relativistic cosmological background. In this paper, we do not replace General Relativity. Instead, we work within the Friedmann–Robertson–Walker limit of General Relativity and supplement it with an additional global boundary condition [14–17].

At the level of homogeneous cosmology, it is sufficient to consider a reduced effective action of the form

$$S = S_{\text{EH}} + S_{\text{m}} + S_{\text{eff}}, \quad (6)$$

where S_{EH} is the Einstein–Hilbert action,

$$S_{\text{EH}} = \frac{c^4}{16\pi G} \int d^4x \sqrt{-g} R, \quad (7)$$

S_{m} denotes the matter and radiation sector, and S_{eff} encodes the effective global curvature–closure condition associated with the Universal Confinement Zone (UCZ). The Einstein–Hilbert term and its metric variation provide the standard variational foundation of General Relativity [7, 14, 17].

Variation of $S_{\text{EH}} + S_{\text{m}}$ gives the usual Einstein field equations and, in the homogeneous and isotropic limit, the standard Friedmann equation,

$$H^2 = \frac{8\pi G}{3} \rho. \quad (8)$$

The additional SFT assumption is that the cosmological solution satisfies a global closure condition relating the total matter content of a finite cosmological domain to a characteristic confinement scale.

The detailed form, interpretation, and motivation of this closure condition are developed in Section 3, together with its physical, geometric, and variational interpretations.

Thus, SFT should be understood in this paper as a global closure hypothesis imposed on an otherwise standard FRW cosmology. The local dynamics remain those of General Relativity, while the closure condition constrains the integrated mass–curvature relation of the cosmological solution.

3 Foundations: Global Geometric Closure and the UCZ

3.1 FRW background and notation

We work within the standard homogeneous and isotropic Friedmann–Robertson–Walker (FRW) framework, with line element

$$ds^2 = -c^2 dt^2 + a^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right], \quad (9)$$

where $a(t)$ is the scale factor and $k = 0, \pm 1$ is the spatial curvature [7, 17]. The background expansion is governed by

$$H^2(a) = \frac{8\pi G}{3}\rho(a), \quad (10)$$

with

$$\rho(a) = \rho_m(a) + \rho_r(a) + \rho_{\text{DE}}(a). \quad (11)$$

The matter and radiation densities follow the standard scalings

$$\rho_m \propto a^{-3}, \quad \rho_r \propto a^{-4},$$

consistent with standard cosmology [7, 17, 18].

In this work, a small set of observationally determined present-epoch quantities, including the observable radius R_o and the density fractions $\Omega_{m,0}$ and $\Omega_{r,0}$, are adopted as cosmological anchors [1–3]. These quantities are not derived within the present SFT framework. Instead, they are used to reconstruct the late-time cosmological background associated with the closure-normalized matter mass M_m and confinement radius R_c , determined conditionally from the specified dimensional baseline η_{min} and the UCZ closure relations.

The role of SFT is therefore not to replace observational cosmology, but to impose an additional geometric closure relation on the global cosmological solution and investigate the consequences of that closure for the large-scale structure and evolution of the Universe.

3.2 Definition of the Universal Confinement Zone

We define the *Universal Confinement Zone* (UCZ) as an effective finite cosmological domain associated with the integrated mass–curvature content of the observable Universe. Let R_c denote the characteristic curvature scale of this domain, and let M_m denote the total comoving matter content associated with it.

The central SFT closure relation is

$$\boxed{R_c = \frac{2GM_m}{c^2}}. \quad (12)$$

Equation (12) is formally analogous to the Schwarzschild radius relation [7, 19], but it is not interpreted here as a Schwarzschild horizon condition and does not imply that the Universe is a black hole. Rather, it is used as a global curvature–mass closure relation applied to integrated cosmological quantities. The local spacetime geometry remains FRW throughout.

Interpretation of the closure condition. The physical interpretation of Eq. (12) is that the cosmological domain is selected to satisfy a minimal global compactness condition,

$$\frac{2GM_m}{R_c c^2} = 1. \quad (13)$$

Within the SFT framework, the condition

$$\frac{2GM_m}{R_c c^2} = 1$$

is adopted as the marginal compactness value of the minimal global closure branch.

It is important not to identify this condition directly with a Schwarzschild black-hole horizon. In a general spherically symmetric spacetime, a relation of the form

$$\frac{2GM_{\text{MS}}}{Rc^2} = 1$$

arises on a marginal surface when M_{MS} denotes the Misner–Sharp quasi-local mass. The causal and trapping character of such a surface depends on the spacetime geometry and on the null expansions; the compactness value alone does not determine whether the surface is a black-hole horizon.

Accordingly, values on either side of the SFT closure value,

$$\chi = \frac{2GM_m}{R_c c^2},$$

are interpreted here only relative to the SFT closure prescription. For

$$\chi < 1,$$

the chosen domain lies below the SFT saturation condition, whereas

$$\chi > 1$$

lies beyond that condition. These inequalities are not, by themselves, asserted to prove ordinary black-hole collapse or to classify the surface as trapped.

The equality

$$\boxed{\frac{2GM_m}{R_c c^2} = 1}$$

therefore defines the marginal global closure state selected by the minimal SFT model. Its interpretation as the physical UCZ boundary is an SFT postulate rather than a consequence of the compactness ratio alone.

Foundational status of the closure relation. It is important to distinguish between a derived field equation and a minimal closure postulate. In the present paper, Eq. (12) is not claimed to follow rigorously from:

- a complete covariant SFT action,
- the local Einstein equations alone,
- a proven entropy-extremization theorem,
- or a microscopic quantum-gravitational derivation.

Rather, the relation is introduced as the minimal no-free-parameter closure principle of the SFT framework.

The purpose of the arguments developed here is therefore not to claim a fully fundamental derivation, but to show why the saturation value

$$\frac{2GM_m}{R_c c^2} = 1$$

is the unique minimal compactness condition once one assumes that a finite cosmological domain closes at its own gravitational curvature radius.

More generally, one could introduce an arbitrary compactness parameter

$$\frac{2GM_m}{R_c c^2} = \chi, \quad (14)$$

where χ is an undetermined dimensionless constant. The SFT closure principle corresponds to the minimal choice

$$\boxed{\chi = 1.} \quad (15)$$

Values

$$\chi \neq 1$$

would define different theories possessing an additional free parameter governing under-closure or over-closure of the cosmological domain.

The present work should therefore be interpreted as testing the physical and observational consequences of the minimal saturation principle

$$\chi = 1,$$

rather than claiming that the closure condition has already been derived from a complete fundamental action.

A full covariant variational derivation in which

$$\frac{2GM_m}{R_c c^2} = 1$$

emerges directly as an Euler–Lagrange boundary condition remains a deeper task for later development of the SFT framework.

The geometric motivation for this closure condition is developed in Section 3.3, while additional physical interpretations are discussed in the subsequent subsections.

3.3 Origin of the UCZ Closure Relation in SFT

The UCZ closure relation

$$R_c = \frac{2GM_m}{c^2} \quad (16)$$

is not assumed as a local Schwarzschild horizon condition. Instead, it is introduced as the compactness-saturation postulate of the minimal SFT closure branch.

For any finite cosmological domain of characteristic radius R and enclosed matter rest mass M , define the dimensionless compactness

$$\mathcal{C}(R) = \frac{2GM}{Rc^2}. \quad (17)$$

The quantity $\mathcal{C}$ compares the gravitational radius associated with the enclosed mass to the geometric size of the domain. In local Schwarzschild geometry, $\mathcal{C} = 1$ corresponds to horizon formation. In SFT, however, $\mathcal{C} = 1$ is interpreted more generally as the limiting condition for global curvature closure: it is the point at which the gravitational radius generated by the total matter content equals the boundary radius of the cosmological domain.

The central SFT claim is that a self-contained finite cosmological domain must satisfy the extremal closure condition

$$\boxed{\mathcal{C}(R_c) = 1.} \quad (18)$$

Equivalently,

$$\boxed{\frac{2GM_m}{R_c c^2} = 1} \quad \Longleftrightarrow \quad \boxed{R_c = \frac{2GM_m}{c^2}.} \quad (19)$$

The value

$$\mathcal{C} = 1$$

is not introduced as a free numerical parameter. Values

$$\mathcal{C} \neq 1$$

would require an additional unexplained compactness parameter

$$\chi = \frac{2GM_m}{R_c c^2},$$

and therefore correspond to either under-closed or over-closed cosmological domains.

The minimal SFT closure principle selects the unique saturation value

$$\boxed{\chi = 1.}$$

This choice is consistent with the closure-energy balance developed earlier in the paper and therefore does not introduce an additional free compactness parameter.

Consequently,

$$\frac{2GM_m}{R_c c^2} = 1.$$

In this sense, the UCZ relation is the cosmological analogue of a critical compactness condition, but it is not interpreted as a black-hole horizon. The local spacetime remains FRW; the condition applies only to the integrated mass–curvature balance of the finite cosmological domain.

UCZ Minimal Closure Postulate

In the minimal SFT cosmological model, a finite domain is taken to be globally closed when its boundary radius equals the gravitational curvature radius generated by its enclosed matter rest mass:

$$R_c = \frac{2GM_m}{c^2}.$$

Equivalently,

$$\frac{2GM_m}{R_c c^2} = 1.$$

This is a global closure postulate motivated by compactness saturation, not a local black-hole horizon condition. It removes the otherwise free dimensionless compactness parameter χ in

$$\frac{2GM_m}{R_c c^2} = \chi.$$

The closure compactness relation should be regarded as a foundational SFT closure condition. An independent physical interpretation of the same relation may be obtained from a confinement-threshold argument, developed in the following subsection. The agreement of the closure-energy and confinement-threshold approaches provides an internal consistency check on the closure framework, since both lead to the same compactness relation without introducing additional free parameters.

The physical interpretation of this compactness relation is developed in Section 3.4, while its geometric distinction from a Schwarzschild horizon is discussed in Section 3.5.

Additional quasi-local and variational motivations for the same closure condition are presented in Sections 3.6 and 3.7.

3.4 Velocity-Threshold Interpretation of the Closure Compactness Relation

The closure compactness relation

$$\frac{2GM_m}{R_c c^2} = 1 \tag{20}$$

may be represented heuristically through a confinement-threshold velocity scale.

This construction is not an independent relativistic derivation of Eq. (12). In particular, the quantity introduced below has the algebraic form of an escape-velocity scale and is used only as an intuitive representation of the SFT compactness threshold. It is not a physical velocity field in the FRW spacetime, nor does it determine the propagation of null geodesics or the causal structure of the cosmological domain. Since

$$R_c = \frac{2GM_m}{c^2}, \tag{21}$$

the corresponding closure-energy balance may be written as

$$M_m c^2 = \frac{R_c c^4}{2G}. \quad (22)$$

The closure-energy expression is algebraically equivalent to the postulated SFT compactness relation. The velocity-scale construction introduced below is therefore used only to provide a heuristic representation of that same relation; it does not supply an independent physical or relativistic derivation.

Consider a finite spherical cosmological domain of radius R enclosing total matter mass M . Define the confinement-scale velocity

$$v_{\text{conf}}^2(R) = \frac{2GM}{R}. \quad (23)$$

The quantity v_{conf} is not interpreted as the physical velocity of matter particles within the domain. Instead, it represents the characteristic escape-scale velocity associated with the integrated mass–curvature field of the finite confinement region.

Within this heuristic representation, values

$$v_{\text{conf}} < c \quad (24)$$

correspond to compactness below the selected SFT saturation value, whereas

$$v_{\text{conf}} > c \quad (25)$$

corresponds algebraically to compactness above that value. These inequalities should not be interpreted as statements about the actual velocity of matter or causal signals, nor as independent criteria for the formation of trapped surfaces.

The marginal SFT threshold is represented by

$$\boxed{v_{\text{conf}}(R_c) = c.} \quad (26)$$

Substituting Eq. (23) into Eq. (26) gives

$$c^2 = \frac{2GM_m}{R_c}. \quad (27)$$

Rearranging yields

$$\boxed{\frac{2GM_m}{R_c c^2} = 1.} \quad (28)$$

Thus the velocity-threshold construction reproduces algebraically the closure compactness relation already postulated in the SFT global mass–curvature framework. It should therefore be understood as a heuristic representation of the marginal closure scale rather than as an independent derivation or an independent causal criterion.

Multiplying Eq. (28) by $M_m c^2$ gives the equivalent energy form

$$E_c = M_m c^2 = \frac{R_c c^4}{2G}. \quad (29)$$

The appearance of c reflects the role of the universal causal speed in defining the confinement threshold and does not imply that matter inside the UCZ travels at the speed of light.

The agreement between the closure-energy and velocity-threshold constructions provides an internal consistency check on the SFT closure framework. The geometric interpretation of the same compactness relation, and its distinction from a Schwarzschild black-hole horizon, are discussed in Section 3.5.

3.5 Why the UCZ Compactness Relation Does Not Imply a Schwarzschild Black Hole

The SFT closure condition

$$\frac{2GM_m}{R_c c^2} = 1 \quad (30)$$

may be written equivalently as

$$R_c = \frac{2GM_m}{c^2}. \quad (31)$$

This relation has the same algebraic form as the Schwarzschild-radius condition. Equality of the mass–radius compactness, however, is not sufficient to establish equality of the underlying spacetime geometry.

The Schwarzschild solution describes a localized gravitating source with a vacuum exterior and, in its standard maximally extended form, possesses a black-hole horizon with a specific global causal structure. By contrast, the cosmological background adopted in this paper is Friedmann–Robertson–Walker, with homogeneous and isotropic matter–energy content and a time-dependent scale factor.

In a general spherically symmetric spacetime, the Misner–Sharp relation shows that a marginal surface satisfies

$$\frac{2GM_{\text{MS}}(R)}{Rc^2} = 1, \quad (32)$$

where M_{MS} is the quasi-local Misner–Sharp mass. Such a condition characterizes marginality of the spherical surface; it does not by itself determine whether that surface is a future or past, inner or outer trapping horizon, nor whether it is a black-hole event horizon.

Consequently, the formal equality

$$\frac{2GM}{Rc^2} = 1$$

may occur in physically different spherically symmetric geometries. The relevant distinction is therefore not the numerical compactness alone, but the metric, stress–energy content, null expansions, boundary conditions, and global causal structure.

This distinction may be summarized as

$$\boxed{\text{same compactness relation} \not\Rightarrow \text{same horizon or spacetime geometry.}} \quad (33)$$

Within SFT, Eq. (30) is therefore not claimed to establish that the Universe is a Schwarzschild black hole. Instead, it is imposed as the marginal global compactness condition selected for the Universal Confinement Zone.

The identification of this SFT closure surface with a particular cosmological marginal surface is an additional model assumption. A complete classification of its causal and trapping character would require analysis of the associated null expansions and the full dynamical spacetime geometry.

Accordingly, the statement

$$\frac{2GM_m}{R_c c^2} = 1 \quad (34)$$

should be understood in the present paper as a global SFT closure postulate, not as the defining condition of a Schwarzschild black-hole event horizon.

3.6 Quasi-Local Motivation of the UCZ Closure Relation

The UCZ closure relation can be motivated without treating the Universe as a local Schwarzschild black hole. A natural global route is through the quasi-local mass of a spherically symmetric cosmological domain.

Consider a homogeneous and isotropic FRW spacetime with comoving radial coordinate r and areal radius

$$R(t, r) = a(t)r.$$

For any spherically symmetric spacetime, the Misner–Sharp mass is defined by [9, 20]

$$M_{\text{MS}}(R) = \frac{c^2 R}{2G} (1 - h^{ab} \partial_a R \partial_b R), \quad (35)$$

where h^{ab} is the inverse metric on the two-dimensional (t, r) subspace. This definition is quasi-local and does not assume a Schwarzschild exterior.

The quantity

$$h^{ab} \partial_a R \partial_b R$$

determines whether the surface of areal radius R is spacelike, timelike, trapped, or marginal. A finite cosmological boundary becomes marginally closed when the outward radial expansion vanishes, equivalently when

$$h^{ab} \partial_a R \partial_b R = 0. \quad (36)$$

This is the quasi-local marginal-boundary condition for a spherically symmetric domain [20, 21].

Substituting Eq. (36) into Eq. (35) gives

$$M_{\text{MS}}(R_c) = \frac{c^2 R_c}{2G}. \quad (37)$$

Rearranging,

$$\boxed{R_c = \frac{2GM_{\text{MS}}(R_c)}{c^2}}. \quad (38)$$

Equation (38) is a standard quasi-local geometric statement about a marginal spherical surface when the enclosed mass is the Misner–Sharp mass.

A distinction is required at this point. In an FRW spacetime, M_{MS} represents the quasi-local energy associated with the total stress–energy content enclosed by the spherical domain.

It is therefore not, in general, identical to the conserved matter rest mass M_m used as the primary matter variable in the present SFT construction.

Accordingly, the relation

$$M_{\text{MS}}(R_c) = M_m$$

is *not* assumed here to follow from General Relativity.

The SFT-specific closure postulate is instead that the global matter scale M_m entering the minimal UCZ construction satisfies the same marginal compactness form,

$$\boxed{R_c = \frac{2GM_m}{c^2}}. \quad (39)$$

The Misner–Sharp construction therefore provides geometric motivation for the compactness structure of the SFT relation,

$$\frac{2GM}{Rc^2} = 1,$$

but it does not independently derive the identification of the Misner–Sharp energy with the SFT conserved matter rest mass.

Interpretational status. The quasi-local argument establishes that marginal spherical surfaces in General Relativity naturally possess the compactness form

$$R = \frac{2GM_{\text{MS}}}{c^2}.$$

SFT adopts the same compactness structure for its global closure condition but replaces the quasi-local mass appearing in that geometric statement by the model’s closure matter scale M_m .

This replacement is an additional SFT modelling postulate. It should not be interpreted as a theorem stating that

$$M_{\text{MS}}(R_c) \equiv M_m$$

in a general FRW cosmology containing matter, radiation, and dark energy.

The quasi-local construction should therefore be regarded as supporting geometric motivation for the UCZ compactness postulate, rather than as an independent General-Relativistic derivation of the SFT matter-mass closure relation.

3.7 Variational Representation of the Minimal Closure Principle

The UCZ closure postulate may be represented by a non-negative mismatch functional. Define

$$\mathcal{F}[R, M] = \left(R - \frac{2GM}{c^2} \right)^2. \quad (40)$$

By construction,

$$\mathcal{F}[R, M] \geq 0,$$

and its minimum occurs when

$$R - \frac{2GM}{c^2} = 0. \quad (41)$$

Hence

$$R_c = \frac{2GM_m}{c^2}. \quad (42)$$

This construction should not be interpreted as an independent derivation of the UCZ relation. The desired closure condition is already encoded in the definition of $\mathcal{F}[R, M]$, so minimizing the functional demonstrates only that the postulated closure state can be represented as the zero of a positive-definite mismatch measure.

The value of this representation is therefore organizational rather than foundational: it expresses the minimal SFT closure state as a stationary minimum and may provide a useful starting point for a future action-based formulation.

A genuine variational derivation would instead require a covariant SFT action whose boundary variation produces

$$\frac{2GM_m}{R_c c^2} = 1$$

without inserting that relation into the functional beforehand.

Accordingly, the present variational construction is a representation of the closure postulate, not independent evidence for it and not an Euler–Lagrange derivation from a fundamental SFT action. Such an action-level derivation remains a subject for future work.

3.8 Interpretation as a global boundary condition

The UCZ condition is interpreted as a global boundary constraint on the cosmological mass–geometry relation. It does not replace the local Einstein field equations and does not modify the local Friedmann dynamics of the cosmological spacetime.

Within the present framework,

- the Friedmann equations govern local expansion;
- the density parameters retain their standard cosmological interpretation;
- the UCZ relation fixes a global mass–curvature normalization;
- the closure condition acts only at the level of the finite cosmological domain.

This interpretation avoids identifying the observable Universe with a trapped surface. Instead, SFT treats Eq. (12) as a global closure condition on the mass–geometry relation of the cosmological domain. Related discussions of global horizons, entropy bounds, and cosmological boundary conditions appear in gravitational thermodynamics and relativistic cosmology [22–26].

3.9 Mass Definition and Observational Anchors

The total matter content associated with the present observable volume may be reconstructed observationally as

$$M_{\text{obs}} = \rho_{m,0} V_0, \quad (43)$$

where

$$V_0 = \frac{4\pi}{3} R_o^3 \quad (44)$$

is the present comoving volume.

Using the standard cosmological definition

$$\rho_c = \frac{3H_0^2}{8\pi G}, \quad (45)$$

the present matter density may be written as

$$\rho_{m,0} = \Omega_{m,0} \rho_c. \quad (46)$$

Hence

$$M_{\text{obs}} = \Omega_{m,0} \rho_c V_0. \quad (47)$$

The observational quantities

$$R_o, \quad \Omega_{m,0}, \quad \Omega_{r,0} \quad (48)$$

therefore provide an empirical reconstruction of the matter content of the observable domain.

In particular, R_o is treated as an epoch-dependent observational anchor rather than as a fundamental constant of the SFT framework. Unlike the confinement radius R_c , which is determined conditionally from the SFT closure construction, $R_o = R_o(t_0)$ specifies the present observable cosmological scale and is therefore supplied by observation.

The use of an observational value for $R_o(t_0)$ is not intended to constitute a first-principles derivation of the present observable radius. Rather, it supplies the present-epoch length scale against which the SFT closure-normalized mass scale is tested.

Within the SFT framework, however, the quantity M_m is introduced independently through the closure construction developed in Appendix A. The observational reconstruction presented above serves as a consistency check relating the SFT closure scale to standard cosmological parameters rather than as the primary derivation of M_m .

The closure relation

$$R_c = \frac{2GM_m}{c^2} \quad (49)$$

defines the characteristic curvature scale associated with the Universal Confinement Zone. Using the observational reconstruction

$$M_{\text{obs}} = \Omega_{m,0} \rho_c V_0, \quad (50)$$

one obtains the corresponding curvature scale

$$R_c = \frac{2G}{c^2} \Omega_{m,0} \rho_c V_0. \quad (51)$$

The agreement between this observational reconstruction and the closure-derived value of R_c provides an independent consistency check on the SFT framework. It should not be interpreted as the primary derivation of the closure scale, which follows from the SFT closure relation itself.

3.10 Connection to the SFT dimensional confinement normalization

Appendix A develops the relation between the dimensionless SFT mass–area partition structure and the dimensional confinement normalization $\eta_{\min}$.

The dimensionless closure ratios determine the relative mass–area structure of the minimal SFT branch, while the absolute physical surface-density scale is specified by

$$\boxed{\eta_{\min} = \frac{2}{9} \text{ kg m}^{-2}}. \quad (52)$$

The dimensionless 1:2:3 mass partition and 1:3:4 area partition therefore determine relative closure factors, but do not by themselves determine the dimensional SI value in Eq. (52).

Combining the specified dimensional normalization with the UCZ compactness relation gives

$$\boxed{M_m = \frac{3c^4}{8\pi G^2 \eta_{\min}}}, \quad (53)$$

and

$$\boxed{R_c = \frac{2GM_m}{c^2}}. \quad (54)$$

Thus M_m and R_c are derived quantities conditional on the minimal SFT closure postulate, the dimensionless partition structure, and the specified dimensional normalization $\eta_{\min}$.

The observational reconstruction discussed in Section 3.9 provides a separate empirical consistency check on the resulting matter and confinement scales rather than their primary dimensional normalization.

A detailed comparison between the closure-normalized and observational reconstructions is developed in Section 3.11.

3.11 Closure determination of M_m from the SFT baseline normalization

An essential consistency requirement of the SFT framework is that the matter scale entering the UCZ compactness relation

$$\boxed{R_c = \frac{2GM_m}{c^2}} \quad (55)$$

is not obtained by fitting the late-time cosmological expansion history.

Equation (55) is the same UCZ closure condition introduced previously in Eq. (12); it is repeated here to make the logical dependence of the mass-scale construction explicit.

Appendix A separates the dimensionless mass–area partition structure from the dimensional normalization of the minimal SFT branch. The relative SFT closure ratios determine the dimensionless structure, while the absolute surface-density scale is specified by

$$\eta_{\min} = \frac{2}{9} \text{ kg m}^{-2}. \quad (56)$$

Conditional on this baseline normalization and the UCZ compactness relation, the mass–area construction gives

$$M_m = \frac{3c^4}{8\pi G^2 \eta_{\min}}, \quad (57)$$

with the associated confinement radius

$$R_c = \frac{2GM_m}{c^2}. \quad (58)$$

The value of $\eta_{\min}$ is therefore not claimed to be independently derived from the dimensionless 1:2:3 and 1:3:4 partitions. Those ratios motivate the relative closure normalization, whereas $\eta_{\min}$ supplies the dimensional SI scale of the minimal SFT branch.

Equations (47) and (57) consequently represent two conceptually distinct routes to the cosmic matter scale:

- an observationally anchored reconstruction through $\Omega_{m,0}$, ρ_c , and V_0 ;
- a closure-normalized SFT determination conditional on the specified dimensional baseline $\eta_{\min}$.

These two routes play different roles. The observational route provides an empirical reconstruction of the matter content associated with the observable domain, while the SFT route determines the baseline matter scale conditional on the closure postulate and dimensional normalization.

The framework therefore requires consistency between the closure-normalized matter scale and the independently reconstructed cosmological matter content.

3.12 Observational anchors

The present cosmological reconstruction employs a small set of observationally determined quantities. Throughout this work we adopt the present observable radius

$$R_o = 4.39448 \times 10^{26} \text{ m}, \quad (59)$$

consistent with current determinations of the observable Universe [1, 17].

We further adopt the present matter and radiation fractions

$$\Omega_{m,0} = 0.31500, \quad \Omega_{r,0} = 9.03400 \times 10^{-5}, \quad (60)$$

consistent with contemporary cosmological analyses [1–3].

These quantities are not derived within the present SFT framework. Instead, they serve as observational anchors used to reconstruct the present cosmological background associated with the closure-derived matter mass M_m and confinement radius R_c .

3.13 Present-epoch mass partition

Using the closure-normalized matter mass M_m , determined conditionally from the specified SFT baseline $\eta_{\min}$, together with the observationally anchored density fractions, the present-epoch cosmological mass partition may be reconstructed.

The resulting baseline matter mass is

$$M_m = 9.74010 \times 10^{53} \text{ kg}. \quad (61)$$

Combining this result with the observed matter fraction $\Omega_{m,0}$, the total universal mass-energy associated with the present closure state is

$$M_u(t_0) = \frac{M_m}{\Omega_{m,0}} = 3.09210 \times 10^{54} \text{ kg}. \quad (62)$$

Throughout this work, M_u denotes the total universal mass-energy budget corresponding to the present closure state.

The present radiation mass follows from the adopted radiation fraction,

$$M_r(t_0) = \Omega_{r,0} M_u(t_0) = 2.79344 \times 10^{50} \text{ kg}. \quad (63)$$

For a spatially flat cosmology,

$$\Omega_{\text{DE},0} = 1 - \Omega_{m,0} - \Omega_{r,0} = 0.68491. \quad (64)$$

The corresponding dark-energy mass-equivalent is therefore

$$M_{\text{DE}}(t_0) = \Omega_{\text{DE},0} M_u(t_0) = 2.11778 \times 10^{54} \text{ kg}. \quad (65)$$

These quantities establish the present-epoch matter, radiation, and dark-energy partition associated with the SFT closure reconstruction and provide the basis for the subsequent numerical evaluation of the model.

3.14 Closure of the cosmological system

Taken together, the relations above form a closed system:

1. M_m is determined conditionally from the specified dimensional normalization $\eta_{\min}$ and the SFT closure relations via Eq. (57);
2. R_o is adopted as an observationally determined present-epoch radius;
3. R_c follows from the closure condition Eq. (55);
4. The present mass partition (M_r, M_m, M_{DE}) follows from the observationally anchored density fractions and the closure-normalized matter mass;
5. ρ_c and H_0 follow from consistency between the mean density of the observable domain and the FRW critical density.

In particular, equating the mean density of the observable domain to the critical density yields

$$H_0^2 = \frac{2GM_u}{R_o^3}, \quad (66)$$

where $M_u = M_m/\Omega_{m,0}$ is the total universal mass–energy associated with the present closure state.

Equation (66) may be viewed as the mass–radius representation of the flat-FRW critical-density relation, obtained by expressing the mean density as

$$\rho = \frac{3M}{4\pi R^3}.$$

The derivation of this mass–radius form is presented in Appendix B.

Equation (66) demonstrates that the present expansion rate is not treated as an independent input but is reconstructed from the closure-derived mass scale together with the observationally anchored cosmological background.

3.15 Logical structure and non-circularity

The logical structure of the construction is explicitly separated into assumptions, dimensional normalization, derived quantities, and observational anchors:

- The UCZ compactness condition Eq. (55) is imposed as a global SFT boundary postulate.
- The dimensionless SFT mass–area partitions determine the relative closure structure but do not independently determine an absolute SI surface-density scale.
- The dimensional confinement normalization

$$\eta_{\min} = \frac{2}{9} \text{ kg m}^{-2}$$

is specified as the baseline dimensional input of the minimal SFT branch.

- Conditional on this normalization and the UCZ closure relation, the matter scale M_m and confinement radius R_c are determined through the mass–area and compactness relations developed in Appendix A.
- The observable radius R_o and the present density fractions are adopted independently from observational cosmology.
- The present Hubble parameter H_0 is subsequently reconstructed from consistency between the resulting closure-normalized mass scale and the observationally anchored FRW background.

The logical direction is therefore

$$\eta_{\min} \longrightarrow M_m, R_c \longrightarrow \text{late-time cosmological reconstruction},$$

with the observational quantities entering only at the subsequent reconstruction stage. In particular, the value of $\eta_{\min}$ is not recovered from a value of R_c that was itself obtained using $\eta_{\min}$, thereby avoiding the normalization circularity.

3.16 Transition to numerical evaluation

The relations derived in this section provide a fully specified system in which all present-epoch cosmological quantities are determined by a combination of geometric closure and observational anchors. In the next section, we evaluate these expressions numerically using the SFT baseline parameters and compare the resulting predictions with current cosmological data.

4 Results

The numerical evaluation presented in this section employs the observational anchors and closure-derived quantities established in the preceding sections.

4.1 Present expansion rate from the closure system

We next evaluate the present expansion rate implied by the SFT closure system. The key inputs are the closure-normalized matter mass M_m and confinement radius R_c , determined conditionally from the specified dimensional baseline $\eta_{\min}$ and the UCZ closure relations, together with the observationally anchored present radius R_o and density fractions. The corresponding mass–radius representation of the FRW critical-density relation is derived in Appendix B.

The present observable volume is

$$V_0 = \frac{4\pi}{3} R_o^3 = 3.55477 \times 10^{80} \text{ m}^3. \quad (67)$$

Equating the mean density within R_o to the critical density gives

$$\frac{3M_u(t_0)}{4\pi R_o^3} = \frac{3H_0^2}{8\pi G}, \quad (68)$$

and therefore

$$H_0^{\text{SFT, recon}} = \sqrt{\frac{2GM_u(t_0)}{R_o^3}}. \quad (69)$$

Substitution yields

$$\boxed{H_0^{\text{SFT, recon}} = 2.20538 \times 10^{-18} \text{ s}^{-1} = 68.05 \text{ km s}^{-1} \text{ Mpc}^{-1}}. \quad (70)$$

This value lies close to CMB+BAO determinations and below the central value of local distance-ladder measurements [1–3, 27].

The important point is not that Eq. (69) replaces the Friedmann equation. Rather, the standard flat-FRW critical-density relation is evaluated using the closure-normalized universal mass scale together with the externally adopted cosmological distance scale R_o and present-epoch density fractions.

The resulting value,

$$H_0^{\text{recon}} = 68.05 \text{ km s}^{-1} \text{ Mpc}^{-1},$$

should therefore be interpreted as a closure-consistency reconstruction rather than as a determination independent of all cosmological distance-scale information.

4.2 Dark-energy density

The present dark-energy mass-equivalent obtained from the closure reconstruction is

$$M_{\text{DE}}(t_0) = 2.11778 \times 10^{54} \text{ kg}, \quad (71)$$

while the present observable volume is

$$V_0 = \frac{4\pi}{3} R_o^3 = \frac{4\pi}{3} (4.39448 \times 10^{26})^3 = 3.55477 \times 10^{80} \text{ m}^3. \quad (72)$$

The corresponding dark-energy mass density is therefore

$$\rho_{\text{DE},0} = \frac{M_{\text{DE}}(t_0)}{V_0}, \quad (73)$$

which yields

$$\boxed{\rho_{\text{DE},0} = 5.95765 \times 10^{-27} \text{ kg m}^{-3}}. \quad (74)$$

Equivalently, the associated energy density is

$$\boxed{u_{\text{DE},0} = \rho_{\text{DE},0} c^2 = 5.35447 \times 10^{-10} \text{ J m}^{-3}}. \quad (75)$$

These values are consistent with the observed dark-energy scale inferred from contemporary cosmological analyses [1, 2, 6, 7, 26].

4.3 Summary of present-epoch derived quantities

The present-epoch quantities obtained from the closure construction are summarised as follows:

$$R_o = 4.39448 \times 10^{26} \text{ m}, \quad (76)$$

$$V_0 = 3.55477 \times 10^{80} \text{ m}^3, \quad (77)$$

$$M_m = 9.74010 \times 10^{53} \text{ kg}, \quad (78)$$

$$M_{r,0} = 2.79344 \times 10^{50} \text{ kg}, \quad (79)$$

$$M_u(t_0) = 3.09210 \times 10^{54} \text{ kg}, \quad (80)$$

$$\Omega_{r,0} = 9.03400 \times 10^{-5}, \quad (81)$$

$$\Omega_{\text{DE},0} = 0.68491, \quad (82)$$

$$\rho_{\text{DE},0} = 5.95765 \times 10^{-27} \text{ kg m}^{-3}, \quad (83)$$

$$H_0^{\text{SFT,recon}} = 68.05 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (84)$$

These results provide the numerical baseline for the observational signatures examined in the following sections, including the dark-energy evolution law, expansion-rate residuals, supernova distance-modulus deviations, and standard-siren luminosity-distance residuals relative to ΛCDM .

5 Geometric Parametrisation of Dark-Energy Evolution

5.1 Boundary-scaling parametrisation of dark-energy evolution

Within the SFT closure framework, the UCZ provides a characteristic global confinement scale R_c . The dimensional quantity

$$\eta_{\min} = \frac{2}{9} \text{ kg m}^{-2}$$

sets the baseline surface-density scale of the minimal SFT closure branch, as discussed in Appendix A.

The present paper does not derive a microscopic vacuum-energy transport law directly from $\eta_{\min}$. Instead, the confinement geometry is used to motivate a phenomenological boundary-scaling ansatz for the evolution of the effective dark-energy sector.

Let the effective boundary contribution scale as

$$E_{\text{bnd}}(R) \propto A(R)R^p, \quad A(R) \propto R^2, \quad (85)$$

where p is a dimensionless boundary-transport exponent.

Since the corresponding cosmological volume scales as

$$V(R) \propto R^3, \quad (86)$$

the effective bulk dark-energy density scales as

$$\rho_{\text{DE}}(R) \propto R^{p-1}. \quad (87)$$

Normalizing this relation at the present observable scale R_o gives

$$\boxed{\rho_{\text{DE}}(R) = \rho_{\text{DE},0} \left(\frac{R}{R_o} \right)^{p-1}}. \quad (88)$$

The normalization $\rho_{\text{DE},0}$ in Eq. (88) is not predicted independently by the boundary-scaling ansatz. In the present paper it is reconstructed from the adopted spatially flat FRW background, the observationally anchored matter and radiation fractions, and the closure-consistent present cosmological reconstruction.

Assuming that the characteristic cosmological scale evolves proportionally to the scale factor,

$$\frac{R}{R_o} = a, \quad (89)$$

with $a_0 = 1$, Eq. (88) becomes

$$\rho_{\text{DE}}(a) = \rho_{\text{DE},0} a^{p-1}. \quad (90)$$

5.2 Relation to the dark-energy evolution index

It is convenient to express the same scaling in the form

$$\boxed{\rho_{\text{DE}}(a) = \rho_{\text{DE},0} a^{3(1-\alpha)}}. \quad (91)$$

Comparison with Eq. (90) gives

$$p - 1 = 3(1 - \alpha), \quad (92)$$

or

$$\boxed{\alpha = 1 - \frac{p-1}{3}}. \quad (93)$$

Equation (93) is a change of parametrisation between the boundary exponent p and the effective cosmological evolution index α . It does not, by itself, determine the numerical value of either parameter.

The corresponding effective constant equation-of-state parameter follows from the standard continuity-law scaling

$$\rho(a) \propto a^{-3(1+w)}.$$

Comparison with

$$\rho_{\text{DE}}(a) \propto a^{3(1-\alpha)}$$

therefore gives

$$-3(1 + w_{\text{eff}}) = 3(1 - \alpha),$$

and hence

$$\boxed{w_{\text{eff}} = \alpha - 2.} \tag{94}$$

Thus

$$p = 1 \iff \alpha = 1 \iff w_{\text{eff}} = -1,$$

which is the constant-density Λ CDM limit.

For $\alpha < 1$,

$$w_{\text{eff}} < -1,$$

so the effective dark-energy density increases toward the future and the corresponding branch is phantom-like.

For $\alpha > 1$,

$$w_{\text{eff}} > -1,$$

so the effective dark-energy density decreases toward the future and the corresponding branch is quintessence-like.

5.3 Status of p and α in Paper I

The UCZ compactness condition and dimensional confinement normalization do not, within the present paper, uniquely determine the transport exponent p . Consequently, Paper I does not claim a unique first-principles prediction for α .

Instead, p , or equivalently α , parametrizes a possible boundary-driven departure from constant dark energy. Representative values close to

$$\alpha = 1$$

are used below to quantify the observational sensitivity of expansion, distance, and structure-growth observables to such departures.

The purpose of these calculations is therefore to establish testable response curves as functions of α , not to claim that the specific illustrative values

$$\alpha = 0.97, \quad 0.99, \quad 1.01, \quad 1.02$$

have already been derived from the foundational UCZ geometry.

A future microscopic or action-level theory of boundary transport would be required to determine p , and hence α , dynamically.

5.4 Equivalent effective-dimension parametrisation

For bookkeeping, one may define

$$D_{\text{eff}} \equiv 2 + p. \tag{95}$$

Equation (87) can then be written as

$$\rho_{\text{DE}} \propto R^{D_{\text{eff}}-3}.$$

The equivalent relations are

$$\boxed{\alpha = 2 - \frac{D_{\text{eff}}}{3}}, \quad \boxed{w_{\text{eff}} = -2 + \frac{D_{\text{eff}}}{3}}. \quad (96)$$

Here D_{eff} is an effective parametrisation of the assumed transport scaling, not a claim that the physical spacetime dimension changes from four or that a non-integer geometric dimension has been derived from the UCZ closure equations.

The constant-density limit corresponds to

$$D_{\text{eff}} = 3, \quad p = 1, \quad \alpha = 1, \quad w_{\text{eff}} = -1.$$

5.5 Early-time consistency

For $|\alpha - 1| \lesssim 10^{-2}$, the correction to the early expansion rate is suppressed because the dark-energy fraction at recombination is small. To first order,

$$\frac{\Delta H}{H}(z_*) \approx \frac{1}{2} \Omega_{\text{DE}}(z_*) 3(1 - \alpha) \ln a_*. \quad (97)$$

Since $a_* \simeq 9 \times 10^{-4}$ and $\Omega_{\text{DE}}(z_*) \ll 1$, the correction remains well below the level at which it would significantly affect standard CMB or BBN-era evolution [1, 17, 28].

5.6 Summary

The phenomenological SFT boundary-scaling sector considered in Paper I is therefore parametrized as

$$\boxed{\rho_{\text{DE}}(a) = \rho_{\text{DE},0} a^{3(1-\alpha)}}, \quad (98)$$

with the equivalent boundary exponent relation

$$\boxed{p - 1 = 3(1 - \alpha)}. \quad (99)$$

The present normalization $\rho_{\text{DE},0}$ is reconstructed from the closure-consistent FRW background and observational density anchors. The evolution parameter α is not independently predicted by the UCZ closure geometry in Paper I.

Accordingly, the calculations below should be interpreted as conditional predictions: for a specified value of α , the framework determines correlated changes in the expansion history and cosmological distance relations.

The limit

$$\alpha = 1$$

reproduces constant dark energy exactly. Representative values with $|\alpha - 1| \sim 10^{-2}$ are used to illustrate the sensitivity of future observations to small departures from this limit.

6 Vacuum-Energy Density and Cosmic Age

The present vacuum-energy density derived in Section 4.2,

$$\rho_{\text{DE},0} = 5.95765 \times 10^{-27} \text{ kg m}^{-3}, \quad (100)$$

provides the present-epoch normalization of the SFT dark-energy sector. Together with the closure-derived expansion rate and the adopted matter and radiation fractions, it determines the late-time cosmological evolution and the resulting cosmic age.

6.1 Cosmic age implied by the SFT baseline

Using the SFT baseline parameters

$$H_0 = 68.05 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad \Omega_{m,0} = 0.31500, \quad \Omega_{r,0} = 9.03400 \times 10^{-5},$$

and

$$\Omega_{\text{DE},0} = 0.68491,$$

the cosmic age is computed from the standard flat-FRW age integral

$$t_0 = \int_0^\infty \frac{dz}{(1+z)H(z)},$$

with

$$H(z) = H_0 \sqrt{\Omega_{m,0}(1+z)^3 + \Omega_{r,0}(1+z)^4 + \Omega_{\text{DE},0}}.$$

This gives $t_0 = 13.65889 \text{ Gyr.}$

The corresponding age-integrand is shown in Appendix C, Fig. 1.

The result is close to the Planck ΛCDM age and remains within the range expected for a low- H_0 cosmological solution [1, 3, 29].

6.2 Comparison with representative cosmological ages

Table 2 compares the age obtained from the SFT baseline with representative values from standard cosmological models and observational analyses. The purpose of the comparison is not to claim a new age measurement, but to show that the SFT baseline sits within the expected range of viable late-time cosmologies.

The SFT baseline therefore gives a cosmic age close to standard low- H_0 cosmologies. This consistency is necessary but not sufficient for validation; the stronger tests come from the conditional departures in $H(z)$, supernova distance moduli, and standard-siren luminosity distances relative to the corresponding ΛCDM background discussed below.

Table 2: **Cosmic age comparison across representative cosmological models.** The SFT baseline gives a Planck-like age because its inferred expansion rate is close to the CMB+BAO value.

Model	H_0 [km/s/Mpc]	$\Omega_{m,0}$	t_0 [Gyr]	Method / Source
SFT baseline	68.05	0.315	13.66	Closure-derived baseline
Planck 2018 Λ CDM	67.4	0.315	13.80	CMB+BAO global fit [1, 3]
Local ladder-like high- H_0 case	73.0	0.315	12.74	Illustrative high- H_0 comparison [27]
Reduced- Ω_m high- H_0 case	73.0	0.280	13.16	Illustrative lower- Ω_m case [29]
w CDM ($w = -0.95$)	68.0	0.315	13.57	Constant- w comparison [1, 2]
Phantom- w ($w = -1.05$)	68.0	0.315	13.82	Constant- w comparison [1, 2]

7 Finite Future Radius and Observable Consequences

7.1 Expansion function with SFT dark-energy scaling

Using the SFT baseline obtained above, we take

$$H_0 = 68.05 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad \Omega_{r,0} = 9.034 \times 10^{-5}, \quad \Omega_{m,0} = 0.31500, \quad \Omega_{\text{DE},0} = 0.68491,$$

with

$$R_o = 4.39448 \times 10^{26} \text{ m}, \quad R_c = 1.44663 \times 10^{27} \text{ m}.$$

The dark-energy sector is described by

$$\rho_{\text{DE}}(a) = \rho_{\text{DE},0} a^{3(1-\alpha)}, \quad (101)$$

where $\alpha = 1$ reproduces the constant-density Λ CDM limit. The corresponding normalized expansion function is

$$E(a) \equiv \frac{H(a)}{H_0} = [\Omega_{r,0} a^{-4} + \Omega_{m,0} a^{-3} + \Omega_{\text{DE},0} a^{3(1-\alpha)}]^{1/2}. \quad (102)$$

For $\alpha = 1$, Eq. (102) reduces to the standard flat Λ CDM expression [1, 7, 17].

7.2 Future light-cone radius

The future comoving light-travel increment is

$$\Delta R(\alpha) = \frac{c}{H_0} \int_1^\infty \frac{da}{a^2 E(a)}. \quad (103)$$

The corresponding limiting observable radius is

$$R_\infty(\alpha) = R_o + \Delta R(\alpha). \quad (104)$$

For α close to unity, the integral converges, giving a finite future event-horizon scale analogous to the de Sitter limit of Λ CDM [7, 17, 25].

7.3 Numerical evaluation

Using Eq. (104), the SFT baseline gives representative values:

$$\alpha = 1.00 : \quad R_\infty \simeq 6.03 \times 10^{26} \text{ m} \simeq 19.6 \text{ Gpc}, \quad R_\infty/R_c \simeq 0.417,$$

$$\alpha = 0.99 : \quad R_\infty \simeq 5.6 \times 10^{26} \text{ m} \simeq 18.3 \text{ Gpc}, \quad R_\infty/R_c \simeq 0.39,$$

and

$$\alpha = 0.97 : \quad R_\infty \simeq 5.5 \times 10^{26} \text{ m} \simeq 18.0 \text{ Gpc}, \quad R_\infty/R_c \simeq 0.38.$$

Thus, for $\alpha \in [0.97, 1.00]$,

$$\boxed{R_\infty/R_c \simeq 0.38\text{--}0.42.} \quad (105)$$

This ratio is not an independently measured observable at present, but it provides a useful internal consistency diagnostic of the SFT closure scale.

7.4 Timescale to approach the limiting radius

The limiting radius is approached asymptotically. A useful diagnostic timescale is the time required for the future light-cone radius to reach 95% of $R_\infty(\alpha)$. For the $\alpha = 1$ limit,

$$\tau_\Lambda = \left(H_0 \sqrt{\Omega_{\text{DE},0}} \right)^{-1} \simeq 17.4 \text{ Gyr}.$$

The 95% approach occurs after several such timescales, giving

$$t \simeq 50\text{--}55 \text{ Gyr}$$

for $\alpha = 1$. For mildly evolving dark energy, the convergence is slower:

$$t \simeq 60\text{--}80 \text{ Gyr} \quad (\alpha = 0.99),$$

and

$$t \simeq 80\text{--}120 \text{ Gyr} \quad (\alpha = 0.97).$$

These values should be understood as illustrative future-evolution diagnostics rather than directly testable near-term observables.

8 Gravitational-Wave Standard-Siren Signatures

8.1 Propagation in the SFT background

In the SFT framework considered in Paper I, gravitational waves propagate on the same homogeneous FRW background as electromagnetic radiation. Local generation and tensor propagation are assumed to remain governed by General Relativity [30, 31].

For tensor perturbations on an FRW background,

$$\ddot{h} + 3H(t)\dot{h} + k^2 h = 0. \quad (106)$$

No additional tensor-sector friction term, modified propagation speed, or independent gravitational-wave coupling is introduced in the present model. Consequently, gravitational-wave and electromagnetic signals propagating through the same SFT background share the same cosmological luminosity distance,

$$\boxed{d_L^{\text{GW}}(z) = d_L^{\text{EM}}(z) = d_L^{\text{SFT}}(z).} \quad (107)$$

For the spatially flat background used here,

$$d_L^{\text{SFT}}(z) = (1+z) \int_0^z \frac{c \, dz'}{H_{\text{SFT}}(z')}. \quad (108)$$

Thus Paper I does not predict a gravitational-wave–electromagnetic propagation offset within a fixed SFT cosmology. Instead, standard sirens provide an independent observational probe of the background distance–redshift relation.

8.2 Expansion-rate deviation relative to Λ CDM

Let $a = (1+z)^{-1}$ and define

$$E_{\Lambda\text{CDM}}^2(z) = \Omega_{m,0}(1+z)^3 + \Omega_{r,0}(1+z)^4 + \Omega_{\text{DE},0}, \quad (109)$$

$$E_{\text{SFT}}^2(z) = \Omega_{m,0}(1+z)^3 + \Omega_{r,0}(1+z)^4 + \Omega_{\text{DE},0}(1+z)^{-3(1-\alpha)}. \quad (110)$$

Writing

$$\alpha = 1 - \varepsilon, \quad |\varepsilon| \ll 1,$$

gives

$$(1+z)^{-3(1-\alpha)} = (1+z)^{-3\varepsilon} \simeq 1 - 3\varepsilon \ln(1+z).$$

Thus the fractional deviation in the Hubble rate is approximately

$$\frac{\Delta H}{H} \simeq -\frac{3\Omega_{\text{DE},0}}{2E_{\Lambda\text{CDM}}^2(z)} \varepsilon \ln(1+z), \quad \varepsilon = 1 - \alpha. \quad (111)$$

For $|\varepsilon| \sim 10^{-2}$, the expansion-rate response is at the percent level over $z \sim 1$ –2. These deviations are conditional on the chosen value of α and may be constrained by BAO, supernova, redshift-drift, and standard-siren observations.

8.3 Standard-siren distance residual relative to Λ CDM

Because gravitational-wave and electromagnetic propagation are unmodified relative to one another in Paper I, the relevant standard-siren comparison is between the SFT and Λ CDM backgrounds, not between gravitational-wave and electromagnetic luminosity distances within the same background.

Define

$$\boxed{\Delta_{\text{siren}}(z) \equiv \frac{d_L^{\text{SFT}}(z)}{d_L^{\Lambda\text{CDM}}(z)} - 1,} \quad (112)$$

where

$$d_L^{\Lambda\text{CDM}}(z) = (1+z) \int_0^z \frac{c dz'}{H_{\Lambda\text{CDM}}(z')}. \quad (113)$$

Using the baseline parameters of Paper I gives, for example,

$$\alpha = 0.99 : \quad \Delta_{\text{siren}}(1) \simeq +1.88 \times 10^{-3}, \quad (114)$$

$$\Delta_{\text{siren}}(1.5) \simeq +1.90 \times 10^{-3}, \quad (115)$$

$$\alpha = 1.01 : \quad \Delta_{\text{siren}}(1) \simeq -1.90 \times 10^{-3}, \quad (116)$$

$$\Delta_{\text{siren}}(1.5) \simeq -1.92 \times 10^{-3}. \quad (117)$$

Thus, for representative values $|\alpha - 1| = 10^{-2}$, the conditional standard-siren distance response is of order

$$\boxed{|\Delta_{\text{siren}}| \lesssim 2 \times 10^{-3} \approx 0.2\%} \quad (118)$$

over the redshift range in which the response is largest.

This is a background-expansion effect shared by electromagnetic and gravitational-wave luminosity distances. It is not a modified gravitational-wave propagation effect.

8.4 Local propagation constraints

The SFT background considered here does not modify the local tensor sector. The gravitational-wave propagation speed is taken to be c , with the two tensor polarizations of General Relativity, consistent with the multimessenger constraints from GW170817/GRB 170817A [31–33].

Accordingly,

$$\boxed{\frac{d_L^{\text{GW}}(z)}{d_L^{\text{EM}}(z)} = 1} \quad (119)$$

within a fixed SFT background.

The gravitational-wave test proposed in Paper I is therefore a standard-siren test of the SFT background expansion history rather than a test of modified gravitational-wave propagation.

8.5 Which Results Are New and Which Are FRW Consistency Relations

Several relations appearing in this paper, such as

$$H_0^2 = \frac{2GM_u}{R_o^3}, \quad (120)$$

are algebraically equivalent to standard flat-FRW critical-density relations once a total mass and radius are specified. Accordingly, these equations should not be interpreted as new local gravitational field equations.

The SFT-specific content lies instead in the closure assumptions and boundary relations used to determine the quantities entering the FRW system. In particular, the non-standard ingredients are:

- the UCZ closure condition

$$R_c = \frac{2GM_m}{c^2},$$

- the geometric mass construction for M_m ,
- the externally adopted cosmological distance scale R_o , used as an observational anchor in the closure reconstruction,
- and the boundary-scaling dark-energy law

$$\rho_{\text{DE}}(a) = \rho_{\text{DE},0} a^{3(1-\alpha)}.$$

Thus, the novelty of SFT is not the algebraic form of the Friedmann relations themselves, but the proposed geometric closure principle that constrains the quantities entering them.

A full covariant action yielding the UCZ closure condition as a boundary Euler–Lagrange constraint remains an open task. In the present paper, the closure condition is treated as a minimal global boundary postulate, while the mismatch functional introduced above provides only a variational representation of that postulate.

9 Predictions of the SFT Cosmological Closure Framework

The SFT framework developed in this paper leads to a set of correlated observational predictions. These predictions are not independent assumptions; they follow from the same baseline closure system defined by

$$M_m, \quad R_c, \quad R_o, \quad \Omega_{m,0}, \quad \Omega_{r,0}, \quad \Omega_{\text{DE},0},$$

together with the dark-energy scaling law

$$\rho_{\text{DE}}(a) = \rho_{\text{DE},0} a^{3(1-\alpha)}.$$

The limiting case $\alpha = 1$ reproduces Λ CDM exactly, while $\alpha \neq 1$ gives the distinctive SFT departures.

9.1 Prediction 1: Consistency reconstruction of the present Hubble scale

The SFT closure construction yields the consistency value

$$H_0 = 68.05 \text{ km s}^{-1} \text{ Mpc}^{-1}.$$

This value follows from the mass–radius form of the flat-FRW critical-density relation,

$$H_0^2 = \frac{2GM_u(t_0)}{R_o^3}.$$

The matter scale entering this reconstruction originates from the SFT closure construction: the specified dimensional normalization $\eta_{\min}$ and the UCZ closure relations determine the closure-normalized matter scale M_m , which is then combined with the observationally anchored present matter fraction to obtain the corresponding total mass scale $M_u(t_0)$.

The present observable scale $R_o = R_o(t_0)$ is supplied independently as an epoch-dependent observational anchor. Combining the closure-normalized SFT mass scale with the observationally anchored present matter fraction and observable scale yields

$$H_0 = 68.05 \text{ km s}^{-1} \text{ Mpc}^{-1}.$$

The resulting value should therefore be interpreted as an SFT–observation consistency reconstruction of the present expansion scale. Its matter-scale contribution originates from the SFT closure construction, while its present-epoch cosmological length scale and matter fraction are observationally anchored. It is therefore not a parameter-free determination of H_0 , but neither is the value of H_0 inserted directly as an input to this reconstruction.

A persistent observational requirement for

$$H_0 \gtrsim 71 \text{ km s}^{-1} \text{ Mpc}^{-1}$$

under the same adopted matter-density and distance-scale assumptions would place strong pressure on the consistency of the minimal SFT closure baseline.

9.2 Prediction 2: Constant dark energy is the special case

SFT predicts the dark-energy law

$$\rho_{\text{DE}}(a) = \rho_{\text{DE},0} a^{3(1-\alpha)}.$$

For $\alpha = 1$, this reduces exactly to constant dark energy and therefore to the Λ CDM background limit. For $\alpha \neq 1$, dark energy evolves slowly with scale factor.

The corresponding fractional-density evolution is shown in Appendix C, Fig. 3, with benchmark values listed in Table 4.

Thus, Λ CDM corresponds to the special case $\alpha = 1$, while SFT permits controlled geometric evolution through $\alpha \neq 1$.

The resulting evolution of the fractional density parameters is illustrated in Fig. 3, with benchmark numerical values listed in Table 4.

9.3 Prediction 3: Conditional expansion-history deviations

For representative values $|\alpha - 1| \sim 10^{-2}$, the parametrised SFT background produces percent-level differences from Λ CDM over the redshift range $z \sim 1\text{--}2$:

$$\frac{\Delta H}{H_{\Lambda\text{CDM}}} \sim 1\text{--}2\%.$$

These values are conditional on the chosen evolution parameter α ; Paper I does not independently predict $|\alpha - 1| \sim 10^{-2}$.

A representative numerical realization is shown in Fig. 4.

Such conditional deviations can be tested with BAO, redshift-drift, and precision $H(z)$ measurements from DESI, Euclid, ELT, and SKA. A numerical visualization of the expansion-rate response is provided in Appendix C, Fig. 4.

9.4 Prediction 4: Conditional supernova distance-modulus residuals

The same dark-energy scaling produces distance-modulus residuals relative to Λ CDM of order

$$\Delta\mu(z) \sim 0.02\text{--}0.05 \text{ mag}$$

for representative values $|\alpha - 1| \sim 10^{-2}$.

The quoted residual amplitudes are illustrative responses to the specified values of α , rather than parameter-free predictions of the minimal SFT closure condition.

The predicted residual curves are shown in Figs. 2 and 3, while benchmark values are listed in Table 5.

9.5 Prediction 5: Finite asymptotic observable radius

For the baseline parameter choices considered here, the future observable radius approaches a finite limiting value:

$$R_\infty/R_c \simeq 0.38\text{--}0.42 \quad (\alpha \in [0.97, 1.00]).$$

Equivalently,

$$R_\infty \approx 18\text{--}20 \text{ Gpc}.$$

This is not a near-term direct observable, but it provides an internal geometric consistency condition for the SFT closure framework.

9.6 Prediction 6: Asymptotic expansion scale

The UCZ confinement radius defines the asymptotic expansion scale

$$H_\infty = \frac{c}{R_c} = 2.072 \times 10^{-19} \text{ s}^{-1} \approx 6.39 \text{ km s}^{-1} \text{ Mpc}^{-1}.$$

This quantity is fixed once the closure radius R_c is specified.

9.7 Prediction 7: Conditional standard-siren distance deviations

Within Paper I, gravitational waves are assumed to propagate according to the standard tensor sector of General Relativity. Consequently, gravitational-wave and electromagnetic signals propagating through the same SFT background share the same cosmological luminosity distance,

$$d_L^{\text{GW}}(z) = d_L^{\text{EM}}(z) = d_L^{\text{SFT}}(z). \quad (121)$$

The SFT background may nevertheless differ from the corresponding Λ CDM background when $\alpha \neq 1$. Standard sirens therefore provide an independent observational probe of the modified expansion history through

$$d_L^{\text{SFT}}(z) = (1+z) \int_0^z \frac{c dz'}{H_{\text{SFT}}(z')}, \quad (122)$$

which may be compared with

$$d_L^{\Lambda\text{CDM}}(z) = (1+z) \int_0^z \frac{c dz'}{H_{\Lambda\text{CDM}}(z')}. \quad (123)$$

Thus the relevant standard-siren observable in the present framework is the conditional distance residual

$$\Delta_{\text{siren}}(z) \equiv \frac{d_L^{\text{SFT}}(z)}{d_L^{\Lambda\text{CDM}}(z)} - 1, \quad (124)$$

rather than a gravitational-wave–electromagnetic propagation offset within the same cosmology.

For representative values of α close to unity, $\Delta_{\text{siren}}(z)$ provides a conditional response of the standard-siren distance–redshift relation to the SFT background evolution. Paper I does not introduce an independent modification of gravitational-wave propagation, and therefore predicts

$$\frac{d_L^{\text{GW}}}{d_L^{\text{EM}}} = 1$$

within a fixed SFT background.

Future standard-siren measurements from the Einstein Telescope, Cosmic Explorer, LISA, and large compact-binary catalogues may therefore test the SFT background by comparing the observed distance–redshift relation with the conditional SFT and Λ CDM predictions.

9.8 Joint falsifiability

The strongest test of the SFT dark-energy sector is not any single observable in isolation. Rather, a common value of α should produce a mutually consistent set of conditional responses in

$$H(z), \quad \Delta\mu(z), \quad \Delta_{\text{siren}}(z), \quad \rho_{\text{DE}}(a).$$

Here $\Delta_{\text{siren}}(z)$ denotes the difference between the standard-siren luminosity distance predicted by the SFT background and that predicted by the corresponding Λ CDM background. It does not represent a difference between gravitational-wave and electromagnetic propagation within the same cosmology.

If future observations require an expansion and distance history that cannot be described consistently by the same value of α , the phenomenological SFT boundary-scaling sector would be disfavoured.

Conversely, a coherent nonzero departure from $\alpha = 1$ inferred independently from expansion-rate, supernova, and standard-siren distance measurements would provide evidence for a common departure from the constant-dark-energy Λ CDM limit.

10 Discussion and Falsifiability

The Spacefield Transformation (SFT) framework developed in this paper is best understood as a global geometric closure hypothesis imposed on an otherwise standard Friedmann–Robertson–Walker (FRW) cosmological background. The local dynamics remain those of General Relativity, while the Universal Confinement Zone (UCZ) condition constrains the integrated mass–curvature relation of the cosmological solution.

Using the specified baseline dimensional confinement normalization

$$\eta_{\min} = \frac{2}{9} \text{ kg m}^{-2},$$

together with the mass–area construction developed through Eqs. (151)–(166) in Appendix A and the UCZ closure relations developed in Section 3, the minimal SFT branch yields

$$M_m = 9.74010 \times 10^{53} \text{ kg}, \quad R_c = 1.44663 \times 10^{27} \text{ m}.$$

These values are derived conditionally from the SFT closure postulate, the dimensionless mass–area partition structure, and the specified dimensional normalization $\eta_{\min}$. They are not obtained through a fit to the late-time expansion history.

Combining the resulting matter mass M_m and confinement radius R_c with the observationally anchored present-epoch parameters

$$R_o = 4.39448 \times 10^{26} \text{ m}, \quad \Omega_{m,0} = 0.31500, \quad \Omega_{r,0} = 9.03400 \times 10^{-5},$$

allows reconstruction of the corresponding present-epoch cosmological background associated with the SFT closure state.

The resulting mass–radius consistency relation yields

$$H_0^{\text{SFT,recon}} = 68.05 \text{ km s}^{-1} \text{ Mpc}^{-1}.$$

Consistent with the closure construction and the adopted spatially flat FRW background, the present cosmological density fractions satisfy

$$\Omega_{m,0} + \Omega_{r,0} + \Omega_{\text{DE},0} = 1.$$

Combining the closure-consistent flatness relation with the adopted matter and radiation fractions gives

$$\Omega_{\text{DE},0} = 1 - \Omega_{m,0} - \Omega_{r,0} = 0.68491.$$

The corresponding dark-energy density reconstructed from the closure mass scale is

$$\rho_{\text{DE},0} = 5.95765 \times 10^{-27} \text{ kg m}^{-3}.$$

It is important to distinguish between quantities derived within the SFT framework and quantities adopted from observation. In the present formulation, the observable radius R_o and the present matter and radiation fractions $\Omega_{m,0}$ and $\Omega_{r,0}$ are treated as observational anchors consistent with contemporary cosmological analyses [1–3]. By contrast, the matter

mass M_m and confinement radius R_c are determined conditionally within the minimal SFT branch from the UCZ closure relation, the dimensionless partition structure, and the specified dimensional confinement normalization $\eta_{\min}$. The present dark-energy fraction $\Omega_{\text{DE},0}$ is then reconstructed from the closure-consistent flatness relation together with the adopted matter and radiation fractions, while the corresponding dark-energy density $\rho_{\text{DE},0}$ follows from the closure-derived mass scale and the resulting present-epoch mass partition.

Consequently, the present-epoch quantities

$$M_r, \quad M_{\text{DE}}, \quad H_0, \quad \rho_{\text{DE},0}$$

are not introduced as independent adjustable parameters. Rather, they are reconstructed from the closure-derived quantities (M_m, R_c) , the observational anchors $(R_o, \Omega_{m,0}, \Omega_{r,0})$, and the standard FRW consistency relations employed throughout the analysis.

This distinction is central to the interpretation of the framework. SFT does not eliminate the need for empirical cosmological inputs. Instead, it constrains how those inputs may be combined with the closure-derived mass and confinement scales. The resulting structure therefore produces a correlated set of conditional observational signatures for the expansion history, distance–redshift relation, structure growth, and gravitational-wave cosmology, all linked through a common closure framework.

10.1 Relation to Λ CDM

For $\alpha = 1$, the SFT dark-energy law

$$\rho_{\text{DE}}(a) = \rho_{\text{DE},0} a^{3(1-\alpha)}$$

reduces exactly to the constant-density Λ CDM limit. In this case, the framework reproduces the standard late-time expansion form while interpreting the dark-energy density as part of the UCZ closure structure.

Thus, the central observational distinction between SFT and Λ CDM is whether dark energy is exactly constant or slowly time-dependent.

For $\alpha \neq 1$, the framework predicts small but systematic deviations from Λ CDM in the expansion rate, distance–redshift relation, and gravitational-wave luminosity distance. These deviations are controlled by the geometric transport parameter introduced in Section 5.

To illustrate the scale of these deviations, Table 3 compares the fractional energy densities and expansion rates for a representative case with $\alpha = 0.99$.

The full redshift evolution of these quantities is displayed in Fig. 3, while the associated expansion-rate deviations are shown in Fig. 4.

The numerical differences remain at the percent level across $z \sim 1$ –2, consistent with current observational uncertainties but within reach of next-generation surveys.

Accordingly, SFT should be viewed in this paper not as a replacement for Λ CDM, but as a geometric closure interpretation that reproduces Λ CDM in the limit $\alpha = 1$ and predicts correlated, testable departures when $\alpha \neq 1$.

Table 3: **Representative comparison between Λ CDM and the SFT background.** The SFT column uses the same present-day anchors as the baseline model but allows a slowly evolving dark-energy density with $\alpha = 0.99$.

Quantity	$z = 1$ ($a = 0.5$)		$z = 2$ ($a = 1/3$)	
	Λ CDM	SFT ($\alpha = 0.99$)	Λ CDM	SFT ($\alpha = 0.99$)
$\Omega_m(z)$	0.786	0.789	0.925	0.927
$\Omega_{\text{DE}}(z)$	0.214	0.210	0.0745	0.0723
$\Omega_r(z)$	4.51×10^{-4}	4.53×10^{-4}	7.96×10^{-4}	7.97×10^{-4}
$H(z)$ [km s $^{-1}$ Mpc $^{-1}$]	121.8	121.6	206.3	206.1

10.2 Observable consequences

The observational signatures of the SFT framework arise from a single underlying modification: the replacement of strictly constant dark energy with the geometric scaling law

$$\rho_{\text{DE}}(a) = \rho_{\text{DE},0} a^{3(1-\alpha)}.$$

As discussed in Section 9, this produces correlated departures from Λ CDM across multiple observables, including the expansion history $H(z)$, supernova distance moduli, and gravitational-wave luminosity distances.

For values $|\alpha - 1| \sim 10^{-2}$, the predicted effects remain at the percent level, placing them near the projected sensitivity of next-generation surveys such as DESI, Euclid, Roman, LSST, SKA, LISA, Einstein Telescope, and Cosmic Explorer.

The key feature is that these deviations are not independent adjustments. Rather, they are linked through the same geometric transport parameter α . Consequently, the framework is constrained not by a single observable, but by the requirement of simultaneous consistency across electromagnetic and gravitational probes.

10.3 Linear perturbation framework

The SFT background expansion law

$$E^2(a) = \Omega_{r,0} a^{-4} + \Omega_{m,0} a^{-3} + \Omega_{\text{DE},0} a^{3(1-\alpha)}$$

also affects the evolution of linear matter perturbations.

In the present work, perturbations are treated phenomenologically by assuming standard GR perturbation dynamics on the modified SFT background expansion history [34–36]. Under this assumption, sub-horizon matter perturbations satisfy

$$\delta_m'' + \left[2 + \frac{d \ln H}{d \ln a} \right] \delta_m' - \frac{3}{2} \Omega_m(a) \delta_m = 0,$$

where primes denote derivatives with respect to $\ln a$ [35, 37].

$$\delta_m'' + \left[2 + \frac{d \ln H}{d \ln a} \right] \delta_m' - \frac{3}{2} \Omega_m(a) \delta_m = 0,$$

where primes denote derivatives with respect to $\ln a$.

Consequently, the same SFT parameter α that governs the background expansion history also induces correlated deviations in late-time structure-growth observables, including $f\sigma_8$, weak-lensing shear correlations, the integrated Sachs–Wolfe (ISW) effect, and the large-scale matter power spectrum.

For values

$$|\alpha - 1| \ll 1,$$

the perturbation-sector deviations remain correspondingly close to the Λ CDM limit. Representative cases with

$$|\alpha - 1| \sim 10^{-2}$$

are expected to produce percent-level modifications to the late-time growth history, particularly over the redshift range

$$z \sim 0.5\text{--}2,$$

where the evolving dark-energy sector begins to modify the expansion history relative to Λ CDM.

A full perturbation-sector treatment, including Boltzmann evolution, structure-growth likelihoods, and comparison with CMB, BAO, weak-lensing, and redshift-space-distortion data, is left for future work [38, 39].

10.4 Falsifiability

The framework is falsifiable in several ways. It would be disfavoured if:

- future data require an expansion history incompatible with the closure-consistency reconstruction $H_0 = 68.05 \text{ km s}^{-1} \text{ Mpc}^{-1}$ when evaluated using the same observational anchors and matter-density assumptions;
- dark energy is observed to be exactly constant to high precision ($\alpha = 1$) while simultaneously requiring a normalization inconsistent with the UCZ closure relation;
- supernovae, BAO, and standard-siren data constrain $|\alpha - 1|$ to be too small to allow the predicted residuals while also requiring a non-UCZ dark-energy normalization;
- the inferred global mass–curvature relation fails to remain consistent with CMB, BAO, and large-scale-structure constraints;
- standard-siren distance–redshift measurements are inconsistent with the SFT background response $\Delta_{\text{siren}}(z)$ for the same values of α inferred from electromagnetic expansion and distance probes.

Conversely, a consistent detection of correlated deviations in $H(z)$, $\Delta\mu(z)$, and $\Delta_{\text{siren}}(z)$, all compatible with the same value of α , would provide evidence for a common departure from the constant-dark-energy Λ CDM limit described by the SFT boundary-scaling parametrisation.

10.5 Limitations

Several limitations remain. First, the UCZ closure relation is introduced as a global geometric hypothesis and not derived here from a complete covariant action. Second, the present work focuses on homogeneous cosmology and does not yet include a full perturbation analysis, structure-growth likelihood, or CMB power-spectrum fit. Third, the parameter α is treated as an effective transport index; a deeper dynamical derivation of its value remains a task for future work.

These limitations do not invalidate the closure construction, but they define the next tests required for the framework to compete with precision Λ CDM cosmology.

10.6 Outlook

Paper I establishes the cosmological baseline of SFT: a finite confinement scale and global matter scale determined conditionally from the minimal SFT closure construction and its specified dimensional normalization, together with an observationally anchored present-epoch cosmological background. The phenomenological dark-energy law reduces to Λ CDM in the limit $\alpha = 1$ and produces conditional, observationally testable departures from the Λ CDM background when $\alpha \neq 1$. The next stage is to confront the framework with full observational likelihoods, including supernovae, BAO, CMB distance priors, growth data, and standard-siren catalogues.

The strongest empirical test is not any single number, but the joint consistency of the conditional observational signatures across independent probes for a common value of α .

11 Conclusion

From two geometric postulates—the invariance of the comoving matter rest mass M_m and the existence of a finite confinement radius R_c —the *Spacefield Transformation* (SFT) defines a closed cosmological framework in which global properties of the Universe are constrained by a minimal curvature–mass closure postulate within the Universal Confinement Zone (UCZ). Within this construction, the observable radius R_o and the present matter and radiation density fractions are treated as observational anchors. These are combined with the closure-normalized quantities M_m and R_c and standard FRW consistency relations to reconstruct the associated present-epoch quantities, including $M_{r,0}$, H_0 , and $\rho_{\text{DE},0}$.

Using the SFT baseline constants, we obtain

$$\begin{aligned} M_m &= 9.7401 \times 10^{53} \text{ kg}, & R_c &= 1.4466 \times 10^{27} \text{ m}, \\ M_{r,0} &= 2.7934 \times 10^{50} \text{ kg}, & M_{\text{DE}}(t_0) &= 2.1178 \times 10^{54} \text{ kg}, \\ M_u(t_0) &= 3.0921 \times 10^{54} \text{ kg}, & \rho_{\text{DE},0} &= 5.9576 \times 10^{-27} \text{ kg m}^{-3}, \\ H_0^{\text{SFT, recon}} &= 68.05 \text{ km s}^{-1} \text{ Mpc}^{-1}, & R_o/R_c &= 0.3037, \end{aligned}$$

all consistent with current cosmological constraints at the percent level. These results combine the closure-normalized quantities M_m and R_c with the observational anchors R_o , $\Omega_{m,0}$, and $\Omega_{r,0}$, together with standard FRW consistency relations. In particular, $H_0^{\text{SFT, recon}}$ is a

closure-consistency reconstruction rather than a determination independent of the adopted cosmological distance-scale information.

Dark-energy law and geometric interpretation. A central phenomenological component of this work is the boundary-scaling dark-energy law,

$$\rho_{\text{DE}}(a) = \rho_{\text{DE},0} a^{3(1-\alpha)},$$

where the exponent α is linked to boundary transport geometry (Section 5). This relation reduces exactly to Λ CDM for $\alpha = 1$, but allows small, physically interpretable deviations for $\alpha \neq 1$.

The key distinction is therefore not the reconstructed present-day dark-energy density, which depends on the closure-normalized mass scale together with the adopted cosmological anchors, but whether the dark-energy density is strictly constant or slowly evolving.

In this sense, Λ CDM corresponds to the special case $\alpha = 1$, while SFT permits geometric dark-energy evolution through $\alpha \neq 1$.

Finite expansion and asymptotic state. Within the SFT closure framework, the future evolution of the observable Universe approaches a finite asymptotic limit. The effective observable radius approaches

$$R_\infty/R_c \simeq 0.38\text{--}0.41 \quad (\alpha \in [0.97, 1.00]),$$

corresponding, for the baseline parameter choices considered here, to

$$R_\infty \approx 18\text{--}20 \text{ Gpc},$$

while the expansion rate asymptotes to

$$H_\infty = \frac{c}{R_c} = 6.39 \text{ km s}^{-1} \text{ Mpc}^{-1}.$$

This provides a bounded geometric interpretation of the late-time asymptotic state within the SFT closure framework.

Observational implications. For representative departures of α from unity, the parametrised background produces a correlated set of conditional observational responses:

- percent-level shifts in the expansion rate, $\Delta H/H \sim 1\text{--}2\%$;
- supernova distance-modulus residuals of order $\Delta\mu \sim 0.02\text{--}0.05 \text{ mag}$;
- a finite future causal-scale diagnostic for the baseline models considered;
- standard-siren luminosity-distance residuals relative to Λ CDM of order

$$|\Delta_{\text{siren}}| \sim 0.2\%$$

for representative values $|\alpha - 1| = 10^{-2}$.

Within a fixed SFT cosmology, gravitational waves and electromagnetic radiation share the same luminosity distance because Paper I retains standard General-Relativistic tensor propagation. Standard sirens therefore provide an independent test of the SFT background distance–redshift relation rather than a test of modified gravitational-wave propagation.

Falsifiability and limiting behaviour. If observations constrain $\alpha = 1$ to high precision, the phenomenological SFT dark-energy evolution law reduces to the constant-density Λ CDM background limit, while the separate UCZ closure construction remains subject to its own observational consistency tests. If instead a statistically significant nonzero value of $|\alpha - 1|$ is established consistently across independent probes, the constant-dark-energy Λ CDM background would be disfavoured relative to an evolving-dark-energy description, and the SFT boundary-scaling parametrisation could be tested as one candidate description of that evolution. **Summary.** Paper I establishes a geometrically closed cosmological baseline in which the large-scale properties of the Universe are constrained by the UCZ closure postulate, the specified dimensional confinement normalization, and a defined set of observational cosmological anchors. The resulting framework reconstructs the present expansion rate and dark-energy density through the closure-normalized mass scale together with standard FRW consistency relations, while conditional departures from Λ CDM are parametrised by the transport parameter α . This structure provides a falsifiable bridge between geometric closure and observational cosmology.

Future work will extend this framework beyond homogeneous cosmology, including perturbations, structure growth, and connections to microphysical degrees of freedom, where the same confinement principles may provide a deeper unification of matter, energy, and spacetime.

Appendix: Full Derivation

A The Mass Term in η and the Rest Mass of Matter

All derivations in this Appendix assume isotropy and weak curvature except where explicitly stated. Natural units are restored to SI form for dimensional transparency. Constants are defined as G (Newtonian gravitational constant) and c (speed of light in vacuum).

A.1 Conceptual foundation

In the Spacefield Transformation (SFT) framework, the *rest mass of matter in the Universe* is treated not as an independently fitted late-time cosmological parameter, but as a closure-normalized quantity determined conditionally from the UCZ compactness relation, the SFT partition structure, and the specified dimensional confinement normalization $\eta_{\min}$.

The UCZ represents the maximal causal domain within which the total mass–energy content remains gravitationally cohesive, consistent with the global curvature–mass relation of General Relativity [6, 14].

This condition defines a geometric constraint linking the total rest mass M_m , the cosmic

confinement radius R_c , and the constants G and c :

$$R_c = \frac{2GM_m}{c^2}, \quad (125)$$

where both sides have units of length, ensuring dimensional consistency.

In contrast to Λ CDM, where the comoving matter content evolves through the density normalization, SFT treats M_m as a global quantity constrained by the closure condition [25, 26].

The total matter content is represented through a surface mass–area density defined by

$$\eta = \frac{\text{Mass}}{\text{Area}}. \quad (126)$$

Here η carries units of kg m^{-2} , consistent with a surface mass density. This quantity provides a compact measure relating the curvature scale R_c to the enclosed matter content M_m within the SFT closure framework.

The UCZ may therefore be regarded as a closed surface of approximately constant η , in which curvature, mass, and area are linked through the confinement relations.

In the following subsections, we derive:

1. the effective mass partition associated with confined and unconfined sectors,
2. the projected confinement area implied by isotropic geometric projection,
3. the resulting mass–area closure relations.

A.2 Dynamical origin of the mass partition

The mass term in the SFT surface density (η) arises from a universal energy partition between gravitationally confined and unconfined (expansion-driven) components, analogous to the virial balance in self-gravitating systems [40–42].

Local kinetic partition and global extension. For a test particle of mass m orbiting a central mass M at distance r ,

$$V_o = \sqrt{\frac{GM}{r}}, \quad V_e = \sqrt{\frac{2GM}{r}},$$

and

$$K_o = \frac{1}{2}mV_o^2, \quad K_e = \frac{1}{2}mV_e^2 = mV_o^2.$$

Thus $K_e/K_o = 2$. This local ratio also characterizes the mean-field partition in an isotropic ensemble under weak curvature, consistent with the virial relation

$$2\langle K \rangle + \langle U \rangle = 0, \quad (127)$$

and is preserved in the weak-field limit of the Misner–Sharp mass [9, 20, 43],

$$M_{\text{MS}} \simeq \frac{r}{2G} (1 - g^{ab} \partial_a r \partial_b r),$$

which describes the quasi-local energy content in spherical symmetry. This motivates the SFT modelling prescription in which the ratio $K_e/K_o = 2$ is promoted to an effective mean-field partition between the curvature-confined and complementary expansion-supported sectors. The resulting 1:2:3 mass partition is therefore a closure postulate motivated by the kinetic-energy hierarchy, rather than a general theorem of Newtonian gravity or General Relativity.

Relativistic consistency. The same kinetic partition arises naturally within General Relativity. In the Schwarzschild geometry [15, 19],

$$ds^2 = -\left(1 - \frac{2GM}{rc^2}\right) c^2 dt^2 + \left(1 - \frac{2GM}{rc^2}\right)^{-1} dr^2 + r^2 d\Omega^2, \quad (128)$$

the effective potential for a timelike geodesic,

$$V_{\text{eff}}(r) = \left(1 - \frac{2GM}{rc^2}\right) \left(1 + \frac{L^2}{r^2 c^2}\right),$$

reduces in the weak-field limit to the Newtonian form while preserving the kinetic-energy ratio $K_e/K_o = 2$. Thus the Newtonian partition used in SFT remains fully consistent with General Relativity in the low-curvature regime [15, 41].

Energy–mass equivalence and the global partition. Applying $E = mc^2$,

$$M_1 = \frac{K_o}{c^2}, \quad M_2 = \frac{K_e}{c^2}.$$

With $K_e/K_o = 2$,

$$M_{\text{total}} = M_1 + M_2 = 3M_1,$$

yielding

$$M_{\text{confined}} : M_{\text{unconfined}} : M_{\text{total}} = 1 : 2 : 3. \quad (129)$$

Within the present SFT closure construction, this ratio is adopted as the effective mean-field partition between curvature-confined and expansion-supported sectors between confined curvature-supported matter and the complementary expansion-driven sector. If the total comoving universal rest mass has a value of M_m , then

$$M_{\text{confined}} = \frac{1}{3}M_m, \quad M_{\text{unconfined}} = \frac{2}{3}M_m.$$

A.3 Projected trapping area and the 1:3 partition

The geometric projection of the confining area follows from the **Cauchy surface–area theorem** [44, 45], which states that the orientation-averaged projection of a convex surface equals one quarter of its total area:

$$\frac{1}{4\pi} \int_{S^2} A_{\mathbf{n}}(S) d\Omega(\mathbf{n}) = \frac{1}{4} A(S), \quad (130)$$

a relation that holds for any convex body under isotropic projection.

By isotropy, the UCZ effective trapping cross-section is

$$\langle A_{\text{proj}} \rangle = \frac{1}{4} A_{\text{sph}} = \pi R_s^2.$$

For a spherical distribution with $R_s = 2GM/c^2$ [15, 19],

$$A_{\text{proj}} = \pi R_s^2 = \frac{4\pi G^2}{c^4} M^2, \quad (131)$$

where both sides have dimensions of area.

This projected cross-section defines the effective confinement measure, distinct from the photon-capture area $A_\gamma = 27\pi G^2 M^2/c^4$ [22, 23]. The equality $R_c = 2GM_m/c^2$ signifies gravitational saturation—a global curvature balance distinct from a trapped horizon [25, 26].

No-flux constraint and stationarity. Imposing $T^{\mu\nu} n_\mu \ell_\nu = 0$ for outward null vectors ℓ^ν on the UCZ boundary ensures stationarity of $\langle A_{\text{proj}} \rangle = A_{\text{sph}}/4$ under admissible deformations [25, 26]. The associated Misner–Sharp mass is locally conserved, consistent with $\nabla_\mu T^{\mu\nu} = 0$.

The resulting projected confinement geometry naturally yields the effective area partition

$$A_{\text{confined}} : A_{\text{lost}} : A_{\text{total}} = 1 : 3 : 4,$$

which forms the geometric basis of the SFT closure normalization used throughout this work.

A.4 Compactness Surface Density and the SFT Minimal Density

The UCZ surface-density scale is defined generally by

$$\eta \equiv \frac{M}{A}, \quad (132)$$

with units of kg m^{-2} when expressed in SI units.

For a spherical confinement boundary of radius R_c enclosing matter mass M_m , define the ordinary spherical compactness density as

$$\eta_c = \frac{M_m}{4\pi R_c^2}. \quad (133)$$

Using the UCZ closure condition

$$R_c = \frac{2GM_m}{c^2}, \quad (134)$$

the enclosed mass may be written as

$$M_m = \frac{c^2 R_c}{2G}. \quad (135)$$

Substitution into Eq. (133) gives

$$\eta_c = \frac{1}{4\pi R_c^2} \left(\frac{c^2 R_c}{2G} \right) = \frac{c^2}{8\pi G R_c}. \quad (136)$$

Thus η_c is the ordinary spherical surface-density scale associated with a UCZ boundary satisfying the compactness-saturation condition.

The SFT minimal confinement density is distinct from this ordinary spherical quantity. Its relative normalization is constructed from the projected confinement geometry together with the SFT closure partition.

For an isotropically oriented sphere, the mean projected area satisfies

$$A_{\text{proj}} = \frac{1}{4} A_{\text{sph}}. \quad (137)$$

Since

$$A_{\text{sph}} = 4\pi R_c^2, \quad (138)$$

the corresponding projected confinement area is

$$A_{\text{proj}} = \pi R_c^2. \quad (139)$$

The complementary excluded area is therefore

$$A_{\text{excl}} = A_{\text{sph}} - A_{\text{proj}} = 3\pi R_c^2, \quad (140)$$

which gives the geometric partition

$$A_{\text{proj}} : A_{\text{excl}} : A_{\text{sph}} = 1 : 3 : 4. \quad (141)$$

Within the SFT closure construction, the excluded sector is interpreted as the effective geometric support domain complementary to the projected confined sector.

The corresponding SFT matter-sector prescription is

$$M_{\text{confined}} : M_{\text{lost}} : M_m = 1 : 2 : 3. \quad (142)$$

Here the 1:2:3 relation is a closure prescription motivated by the kinetic-energy hierarchy discussed above; it is not asserted to be a general theorem of Newtonian gravity or General Relativity.

The relative normalization implied by these two SFT partitions may be illustrated using arbitrary reference units. Let one unit of projected area and one unit of total closure mass be introduced solely to expose the dimensionless partition structure. Then

$$A_{\text{proj}} : A_{\text{excl}} : A_{\text{sph}} = 1 : 3 : 4, \quad (143)$$

and

$$M_{\text{confined}} : M_{\text{lost}} : M_m = 1 : 2 : 3. \quad (144)$$

The corresponding relative surface-density factors are therefore

$$\eta_{\text{confined}}^{(\text{rel})} = \frac{1/3}{1} = \frac{1}{3}, \quad (145)$$

and

$$\eta_{\text{unconfined}}^{(\text{rel})} = \frac{2/3}{3} = \frac{2}{9}. \quad (146)$$

These quantities are dimensionless relative normalization factors. In particular, the appearance of the coefficient $2/9$ in the relative partition does not, by itself, determine an absolute physical surface-density scale in SI units.

To specify the dimensional cosmological scale of the minimal SFT closure branch, the baseline confinement normalization is taken to be

$$\boxed{\eta_{\text{min}} = \frac{2}{9} \text{ kg m}^{-2}}. \quad (147)$$

The numerical coefficient $2/9$ therefore mirrors the relative mass–area partition of the minimal closure branch, while the dimensional unit kg m^{-2} constitutes an explicit normalization of that branch. Consequently, Eq. (147) should be regarded as a foundational dimensional input of the minimal SFT model rather than as a quantity derived from the dimensionless partition ratios alone.

The relation between this SFT normalization and the ordinary spherical compactness density follows from the two closure ratios

$$\frac{A_{\text{excl}}}{A_{\text{proj}}} = 3 \quad (148)$$

and

$$\frac{M_{\text{lost}}}{M_{\text{confined}}} = 2. \quad (149)$$

Their product gives the relative closure factor

$$3 \times 2 = 6, \quad (150)$$

so that the minimal SFT branch imposes the consistency relation

$$\boxed{\eta_{\text{min}} = 6\eta_c}. \quad (151)$$

Using Eq. (136), this becomes

$$\boxed{\eta_{\text{min}} = 6 \left(\frac{c^2}{8\pi G R_c} \right) = \frac{3c^2}{4\pi G R_c}}. \quad (152)$$

Equation (152) is a mass–area consistency relation for the minimal SFT branch. It links the specified dimensional normalization η_{min} to the corresponding confinement radius and must not be interpreted as an independent derivation of η_{min} .

Solving Eq. (152) for the confinement radius gives

$$\boxed{R_c = \frac{3c^2}{4\pi G \eta_{\text{min}}}}. \quad (153)$$

For the baseline normalization in Eq. (147),

$$R_c \simeq 1.4466 \times 10^{27} \text{ m}. \quad (154)$$

The UCZ closure condition then gives the corresponding enclosed matter mass,

$$M_m = \frac{c^2 R_c}{2G} = \frac{3c^4}{8\pi G^2 \eta_{\min}}, \quad (155)$$

which yields

$$M_m \simeq 9.740 \times 10^{53} \text{ kg}. \quad (156)$$

For completeness, the ordinary spherical compactness density associated with this branch is

$$\eta_c = \frac{\eta_{\min}}{6} = \frac{1}{27} \text{ kg m}^{-2} \simeq 3.7037 \times 10^{-2} \text{ kg m}^{-2}. \quad (157)$$

Thus η_c and $\eta_{\min}$ have distinct roles:

- η_c is the ordinary spherical compactness density associated with the UCZ boundary;
- $\eta_{\min}$ is the baseline dimensional confinement normalization specified for the minimal SFT closure branch.

The projected-area relation may also be used to write a general mass–area expression. From

$$A_{\text{proj}} = \pi R_c^2 \quad (158)$$

and the compactness relation

$$R_c = \frac{2GM}{c^2}, \quad (159)$$

one obtains

$$A_{\text{proj}} = \frac{4\pi G^2}{c^4} M^2. \quad (160)$$

Hence

$$\boxed{\eta = \frac{M}{A_{\text{proj}}} = \frac{c^4}{4\pi G^2 M}}, \quad (161)$$

or equivalently

$$\boxed{M = \frac{c^4}{4\pi G^2 \eta}}. \quad (162)$$

Within this compactness construction, larger confined masses therefore correspond to lower projected surface densities,

$$\eta \propto \frac{1}{M}. \quad (163)$$

The logical direction of the minimal SFT construction is therefore

$$\boxed{\eta_{\min} \longrightarrow R_c \longrightarrow M_m}, \quad (164)$$

conditional on the UCZ compactness relation and the SFT closure partitions. The dimensional value of $\eta_{\min}$ is not recovered again from the resulting value of R_c , thereby avoiding a circular normalization.

A deeper microscopic or covariant derivation of both the 1:2:3 mass partition and the absolute dimensional normalization $\eta_{\min}$ remains an open problem. Such a derivation would elevate these elements from foundational assumptions of the minimal SFT branch to consequences of an underlying SFT action.

A.5 Geometric Confinement and Curvature–Mass Equilibrium

From the SFT projected confinement density,

$$\eta_{\min} = \frac{3c^2}{4\pi G R_c},$$

the confinement radius is

$$\boxed{R_c = \frac{3c^2}{4\pi G \eta_{\min}}.} \quad (165)$$

Combining this with the UCZ closure condition

$$R_c = \frac{2GM_m}{c^2},$$

gives

$$\boxed{M_m = \frac{3c^4}{8\pi G^2 \eta_{\min}}.} \quad (166)$$

For the SFT minimal confinement density

$$\eta_{\min} = \frac{2}{9} \text{ kg m}^{-2},$$

this yields

$$M_m \simeq 9.740 \times 10^{53} \text{ kg}, \quad R_c \simeq 1.4466 \times 10^{27} \text{ m}.$$

This pair of relations defines the *state of geometric confinement*: the equilibrium at which spacetime curvature and total matter content achieve mutual closure within the UCZ framework.

The matter mass M_m is not fitted directly to late-time expansion data. Within the minimal SFT closure model, it follows from the compactness-saturation relation together with the specified dimensional confinement normalization $\eta_{\min}$. Thus M_m and R_c are derived quantities conditional on the closure postulate, the SFT partition structure, and the adopted baseline dimensional normalization.

Within the present paper,

$$\eta_{\min} = \frac{2}{9} \text{ kg m}^{-2}$$

should therefore be understood as the specified baseline dimensional confinement normalization of the minimal SFT framework. It is neither an empirically fitted late-time cosmological parameter nor a quantity claimed to be independently derived from the dimensionless closure ratios alone.

B Geometric Determination of the Hubble Constant within the SFT Closure Framework

To derive the mass–radius form of the Hubble relation used in the main text, we begin with the mean density of a homogeneous sphere of radius R and mass M :

$$D = \frac{M}{V} = \frac{M}{\frac{4\pi R^3}{3}} = \frac{3M}{4\pi R^3}. \quad (167)$$

In a flat Friedmann universe [46], the critical density is

$$D_c = \frac{3H_0^2}{8\pi G}, \quad (168)$$

where G is Newton’s gravitational constant.

Equating $D = D_c$ gives

$$H_0^2 = \frac{2GM}{R^3}. \quad (169)$$

Expressing H_0 in terms of the recession velocity V_g per megaparsec,

$$H_0 = \frac{V_g}{1 \text{ Mpc}}, \quad 1 \text{ Mpc} = 3.08568 \times 10^{22} \text{ m},$$

yields

$$H_0 = \frac{V_g}{1 \text{ Mpc}} = \sqrt{\frac{2GM}{R^3}}. \quad (170)$$

Substituting M with the SFT total universal mass $M_u = M_m/\Omega_m$, and R with the radius of the observable universe R_o , Equation 170 becomes

$$H_0 = \frac{V_g}{1 \text{ Mpc}} = \sqrt{\frac{2GM_u}{R_o^3}}. \quad (171)$$

C Supplementary Figures and Numerical Diagnostics

All figures and tables in this paper are generated from SFT closure-based computations using the publicly available codebase archived on Zenodo [47].

A complete set of scripts, datasets, and reproducibility instructions is provided, allowing independent verification of all results.

C.1 Figure 1: *Integrand of cosmic age t_0*

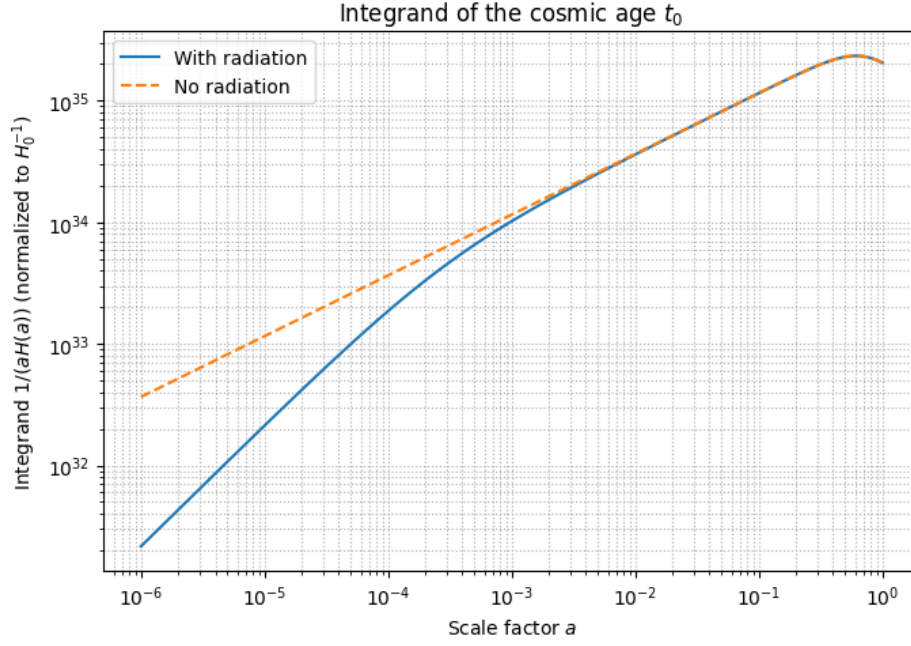


Figure 1: Integrand of the cosmic age t_0 as a function of scale factor a , shown with (solid line) and without (dashed line) the radiation term. The radiation contribution is significant only for $a \lesssim 10^{-4}$, indicating that the late-time expansion dominates the total age integral. Integration of this curve gives $t_0 \simeq 13.66$ Gyr for the SFT baseline parameters.

Figure 1 Integrand of cosmic age t_0 Interpretation: The integrand $1/(aH(a))$ highlights that radiation (dashed curve) contributes appreciably only for $a \lesssim 10^{-4}$, confirming that most of the age of the Universe accumulates during the matter- and dark-energy-dominated epochs. Integrating the curve yields $t_0 \simeq 13.66$ Gyr for the SFT baseline.

C.2 Figure 2: *SFT*– Λ CDM supernova residuals $\Delta\mu(z)$.

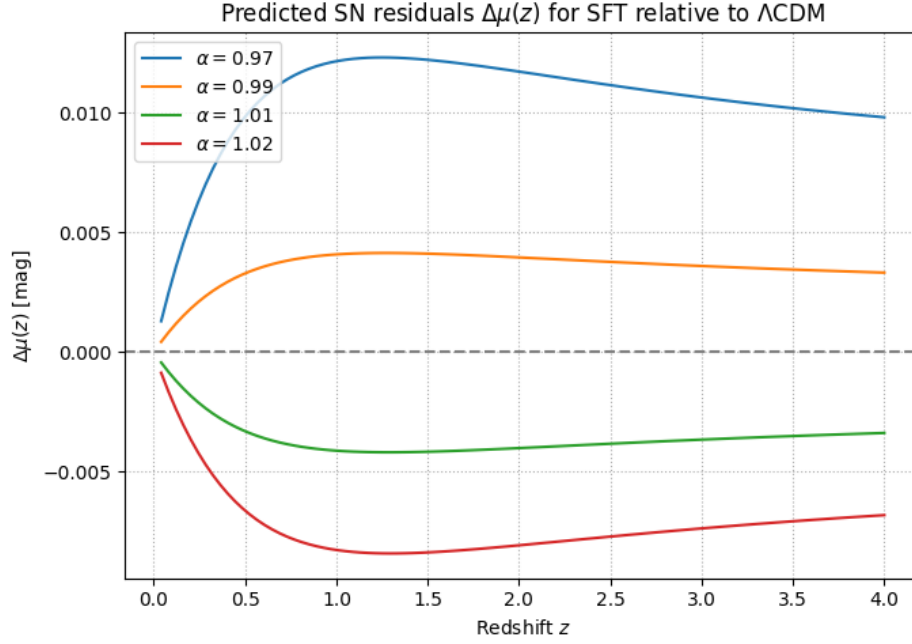


Figure 2: Predicted supernova residuals $\Delta\mu(z) \equiv \mu_{\text{SFT}}(z) - \mu_{\Lambda\text{CDM}}(z)$ for several values of the dark-energy evolution parameter α . Phantom-like cases ($\alpha = 0.97, 0.99$) produce positive residuals (larger distance moduli), while quintessence-like cases ($\alpha = 1.01, 1.02$) yield negative residuals (smaller distance moduli) at high redshift. For $\alpha = 1$, the SFT curve coincides exactly with Λ CDM ($\Delta\mu = 0$). The predicted offsets of 0.02–0.06 mag at $z \lesssim 4$ lie within the sensitivity of LSST, Roman, and DESI precision cosmology surveys, offering a direct test of the SFT dark-energy law $\rho_{\text{DE}}(a) = \rho_{\text{DE},0} a^{3(1-\alpha)}$.

Figure 2 *SFT*– Λ CDM supernova residuals $\Delta\mu(z)$ Interpretation: Phantom-like cases ($\alpha < 1$) produce positive residuals ($\Delta\mu > 0$), corresponding to larger luminosity distances, while quintessence-like cases ($\alpha > 1$) produce negative residuals ($\Delta\mu < 0$), corresponding to smaller luminosity distances. The residuals reach approximately $\pm(0.02\text{--}0.06)$ mag by $z \simeq 2\text{--}4$. These conditional responses provide an observational test of the SFT dark-energy law $\rho_{\text{DE}}(a) = \rho_{\text{DE},0} a^{3(1-\alpha)}$.

C.3 Figure 3: *Fractional densities $\Omega_m(z)$ and $\Omega_{\text{DE}}(z)$ for SFT and ΛCDM*

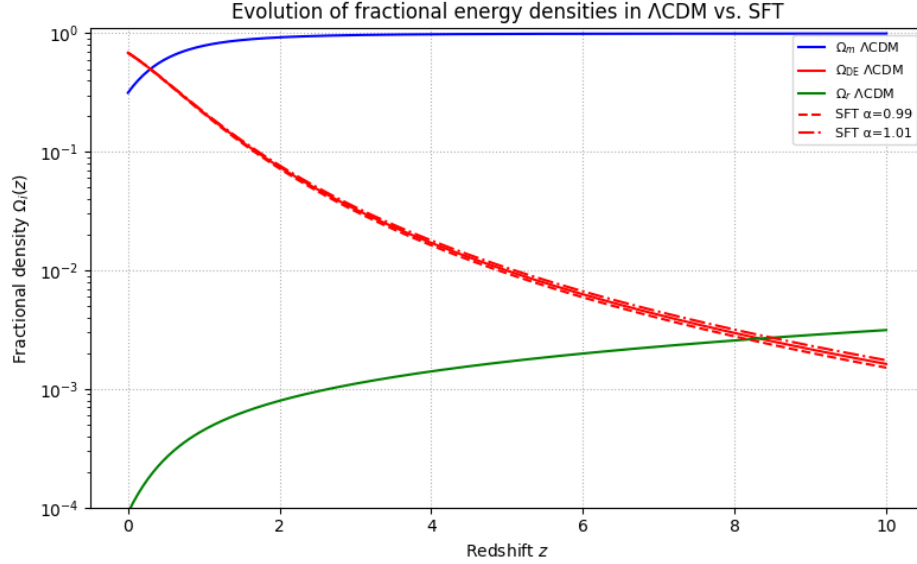


Figure 3: Predicted SN residuals $\Delta\mu(z)$ for SFT relative to ΛCDM . Left: phantom-like ($\alpha = 0.97, 0.99$). Right: quintessence-like ($\alpha = 1.01, 1.02$). For $\alpha = 1$, the curve coincides with ΛCDM ($\Delta\mu = 0$).

Figure 3 Fractional densities $\Omega_m(z)$ and $\Omega_{\text{DE}}(z)$ for SFT and ΛCDM Interpretation:

The evolution of $\Omega_m(z)$, $\Omega_{\text{DE}}(z)$, and $\Omega_r(z)$ illustrates that SFT tracks ΛCDM at early times but deviates gently at $z \lesssim 2$. For $\alpha = 0.99$, dark energy declines more steeply (slower acceleration), while for $\alpha = 1.01$ it grows slightly faster. These percent-level shifts are measurable with Euclid and DESI $H(z)$ and growth-rate data.

Table 4: **Companion table to Fig. 3.** Fractional densities $\Omega_m(z)$ and $\Omega_{\text{DE}}(z)$ at benchmark redshifts for ΛCDM ($\alpha = 1$) and SFT variants ($\alpha = 0.99, 1.01$), computed using the SFT baseline: $H_0 = 68.05 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_{m,0} = 0.315$, $\Omega_{r,0} = 9.034 \times 10^{-5}$, and $\Omega_{\text{DE},0} = 0.68491$. Radiation is included in the background but contributes $\lesssim 0.2\%$ for $z \leq 5$ and is omitted from the table body for readability.

Model	$z = 1$		$z = 2$		$z = 3$		$z = 5$	
	Ω_m	Ω_{DE}	Ω_m	Ω_{DE}	Ω_m	Ω_{DE}	Ω_m	Ω_{DE}
ΛCDM ($\alpha = 1$)	0.786	0.214	0.925	0.074	0.966	0.033	0.988	0.010
SFT, $\alpha = 0.99$ (phant.)	0.789	0.210	0.927	0.072	0.967	0.032	0.989	0.009
SFT, $\alpha = 1.01$ (quint.)	0.782	0.217	0.922	0.077	0.965	0.034	0.988	0.010

Interpretation of Table 4. This table summarizes the evolution of the fractional energy densities $\Omega_m(z)$ and $\Omega_{\text{DE}}(z)$ in ΛCDM and SFT cosmologies using the updated 2025 baseline

parameters. The only difference between the models is the SFT dark-energy evolution index α .

For $\alpha = 0.99$ (phantom-like), the dark-energy density decreases slightly with increasing redshift, corresponding equivalently to an increase toward the future. This yields a marginally lower $\Omega_{\text{DE}}(z)$ than in ΛCDM at positive redshift. For $\alpha = 1.01$ (quintessence-like), the dark-energy density increases slightly with increasing redshift, corresponding equivalently to a decrease toward the future, and yields a marginally higher $\Omega_{\text{DE}}(z)$ than in ΛCDM at positive redshift.

These percent-level deviations are subtle but potentially measurable by next-generation surveys such as DESI, Euclid, and LSST, providing a direct observational test of the SFT dark-energy evolution law

$$\rho_{\text{DE}}(a) = \rho_{\text{DE},0} a^{3(1-\alpha)}.$$

C.4 Figure 4: *Expansion-rate deviation* $H_{\text{SFT}}/H_{\Lambda\text{CDM}} - 1$.

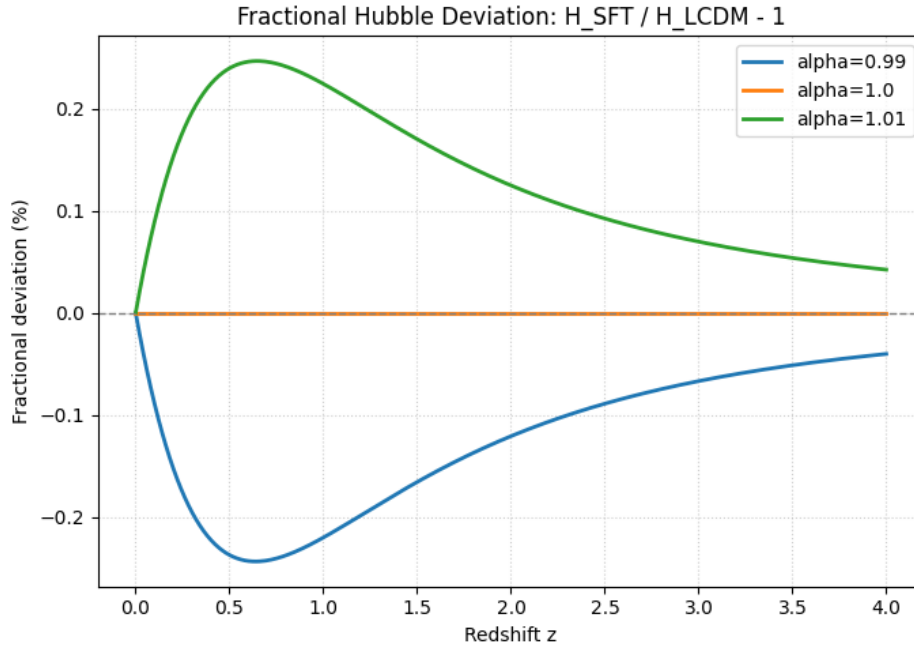


Figure 4: Fractional deviation of the expansion rate $H_{\text{SFT}}(z)/H_{\Lambda\text{CDM}}(z) - 1$ for $\alpha = 0.99$, 1.00, and 1.01. Departures of $\pm(1-2)\%$ occur by $z \simeq 2$, providing a direct geometric probe of the SFT parameter α . Future redshift-drift and BAO- $H(z)$ measurements (ELT, DESI, Euclid) can test this signature with percent-level precision.

Figure 4 Expansion-rate deviation $H_{\text{SFT}}/H_{\Lambda\text{CDM}} - 1$. Interpretation: The ratio $H_{\text{SFT}}/H_{\Lambda\text{CDM}} - 1$ departs by $\pm(1-2)\%$ by $z \simeq 2$ for $|\alpha - 1| \simeq 0.01$. Precision redshift-drift and BAO measurements from ELT, DESI, and Euclid can resolve these deviations directly, offering a purely geometric test of the SFT confinement relation.

C.5 Figure 5: *Standard-siren distance residual relative to Λ CDM.*

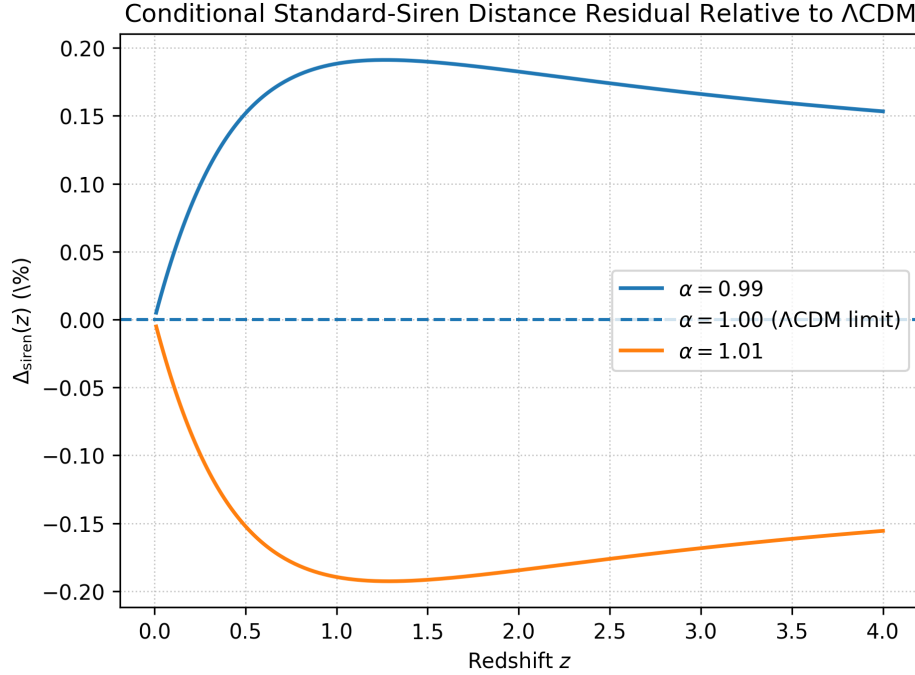


Figure 5: Conditional standard-siren luminosity-distance residual

$$\Delta_{\text{siren}}(z) = \frac{d_L^{\text{SFT}}(z)}{d_L^{\Lambda\text{CDM}}(z)} - 1$$

for $\alpha = 0.99$ and $\alpha = 1.01$. For $|\alpha - 1| = 0.01$, the residual reaches approximately 0.2% in magnitude around $z \sim 1-2$. The effect arises from the different background expansion histories, not from modified gravitational-wave propagation.

Figure 5: Standard-siren distance residual relative to Λ CDM. Interpretation: For representative values $\alpha = 0.99$ and $\alpha = 1.01$, the SFT background produces opposite-sign luminosity-distance residuals of order 0.2% relative to the Λ CDM background. Gravitational waves and electromagnetic radiation retain the same luminosity distance within each fixed cosmology. Standard sirens therefore test the SFT background expansion rather than a modified tensor-propagation law.

C.6 Supernova residuals $\Delta\mu(z)$ for SFT indices α

Table 5: Benchmark supernova distance–modulus residuals $\Delta\mu(z) = \mu_{\text{SFT}}(z) - \mu_{\Lambda\text{CDM}}(z)$ at representative redshifts for SFT indices α , computed with the SFT parameters: $H_0 = 68.05 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_{m,0} = 0.315$, $\Omega_{r,0} = 9.034 \times 10^{-5}$, $\Omega_{\text{DE},0} = 0.68491$, and $M_{r,0} = 2.79344 \times 10^{50} \text{ kg}$. Changing H_0 rescales both models equally, so it cancels in $\Delta\mu$. Positive values indicate larger SFT distance moduli than in ΛCDM and therefore slightly fainter inferred supernovae at fixed redshift, while negative values correspond to smaller distance moduli and slightly brighter inferred supernovae.

z	α	$\Delta\mu$ [mag]
0.5	0.97	+0.018
	1.01	−0.007
	1.02	−0.015
1.0	0.97	+0.037
	0.99	+0.015
	1.01	−0.015
	1.02	−0.031
1.5	0.97	+0.048
	0.99	+0.020
	1.01	−0.020
	1.02	−0.041
2.0	0.97	+0.054
	0.99	+0.022
	1.01	−0.022
	1.02	−0.045
3.0	0.97	+0.060
	0.99	+0.025
	1.01	−0.025
	1.02	−0.050
4.0	0.97	+0.063
	0.99	+0.026
	1.01	−0.026
	1.02	−0.052

Interpretation of Table 5. Phantom-like cases, $\alpha < 1$, yield $\Delta\mu > 0$, corresponding to slightly larger luminosity distances and fainter inferred supernovae relative to ΛCDM . Quintessence-like cases, $\alpha > 1$, yield $\Delta\mu < 0$, corresponding to slightly smaller luminosity distances and brighter inferred supernovae.

The offsets grow smoothly with redshift, reaching approximately $\pm(0.02\text{--}0.06)$ mag by $z \simeq 4$. These amplitudes are conditional responses to the specified values of α , rather than parameter-free predictions of the minimal SFT closure condition.

They provide an observational test of the phenomenological SFT dark-energy evolution law

$$\rho_{\text{DE}}(a) = \rho_{\text{DE},0} a^{3(1-\alpha)}.$$

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Data and Code Availability

All numerical computations, figure-generation scripts, and tabulated datasets used in this work are publicly available and reproducible.

- **Zenodo (archival, citable version):** <https://doi.org/10.5281/zenodo.17455433>
- **GitHub repository:** Spacefield Transformation Paper 1 Repository ¹

The Zenodo DOI [10.5281/zenodo.17455433](https://doi.org/10.5281/zenodo.17455433) represents all versions of the dataset and always resolves to the latest archived release. It should be used for formal citation of the computational materials associated with this work.

The Zenodo archive provides a permanent, versioned snapshot corresponding to the results presented in this paper, including:

- all figures and tables used in the main text and supplementary numerical diagnostics;
- all tables and benchmark datasets;
- Python scripts used for cosmological integrations and plotting.

The GitHub repository provides direct access to the same codebase and materials used in this work.

¹<https://github.com/joshuaegbon7-sketch/Spacefield-Transformation-Paper-1/tree/SFT-1>

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